

Evolution of olfactory sensitivity, preferences and behavioral responses in Mexican cavefish: fish personality matters

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Abstract

Animals are adapted to their natural habitats and lifestyles. Their brains perceive the external world *via* their sensory systems, compute information together with that of internal states and autonomous activity, and generate appropriate behavioral outputs. However, how do these processes evolve across evolution? Here, focusing on the sense of olfaction, we have studied the evolution in olfactory sensitivity, preferences and behavioral responses to six different food-related amino acid odors in the two eco-morphs of the fish *Astyanax mexicanus*. To this end, we have developed a high-throughput behavioral setup and pipeline of quantitative and qualitative behavior analysis, and we have tested 489 six-week-old *Astyanax* larvae. The blind, dark-adapted morphs of the species showed markedly distinct basal swimming patterns and behavioral responses to odors, higher olfactory sensitivity and a strong preference for alanine, as compared to their river-dwelling eyed conspecifics. In addition, we discovered that fish have an individual “swimming personality”, and that this personality influences their capability to respond efficiently to odors and find the source. Importantly, the personality traits that favored significant responses to odors were different in surface fish and cavefish. Moreover, the responses displayed by second-generation cave x surface F2 hybrids suggested that olfactory-driven behavior and olfactory sensitivity is a quantitative genetic trait. Our findings show that olfactory processing has rapidly evolved in cavefish at several levels: detection threshold, odor preference, and foraging behavior strategy. Cavefish is therefore an outstanding model to understand the genetic, molecular and neurophysiological basis of sensory specialization in response to environmental change.

eLife assessment

In this **important** paper, Blin and colleagues develop a high-throughput behavioral assay to test spontaneous swimming and olfactory preference in individual Mexican cavefish larvae. The authors present **compelling** evidence that the surface and cave morphs of the fish show different olfactory preferences and odor sensitivities and that individual fish show substantial variability in their spontaneous activity that is relevant for olfactory behaviour. The paper will be of interest to neurobiologists working on the evolution of behaviour, olfaction, and the individuality of behaviour.

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Introduction

With more than 26,000 species representing half of vertebrates, bony fishes are extremely diverse and have colonized all possible ecological niches (Helfman et al., 2009 [↗](#)), making them outstanding models to study the neurophysiological, genetic and evolutionary underpinnings of adaptive behaviors. To colonize, survive and thrive in various environments, sensory systems are particularly crucial as they serve as windows to the external world, and they display an exceptional diversity in fishes. From the visual system of deep-sea fish shaped to catch the rare photons or the point-like bioluminescent signals to the mobile barbels covered with taste buds to probe the sea bottom and detect buried prey of bottom feeders, examples abound (de Busserolles et al., 2020 [↗](#); Kiyohara et al., 2002 [↗](#)).

The anatomical sensory specializations, that is the relative importance taken by specific sensory organs or brain sensory areas, arise during embryonic development through regulatory processes that control the patterning of the neuroepithelium and that define boundaries between presumptive neural regions (Krubitzer et al., 2011 [↗](#)). The relative investment in different sensory modules governs anatomical and behavioral specializations. Among fish, sand-dwelling cichlids that rely almost exclusively on vision to execute behaviors develop a large optic tectum, whereas rock-dwelling cichlids, which inhabit complex environments, have small optic tecta but an enlarged telencephalon, and these variations arise early in development (Sylvester et al., 2010 [↗](#)). Similarly, in the embryos of dark-adapted blind Mexican cavefish, the presumptive eyefield territory is reduced but the presumptive olfactory epithelium is increased in size, presumably as a compensation for the loss of visual modality in the dark environment (Agnès et al., 2022 [↗](#); Hinaux et al., 2016a [↗](#); Pottin et al., 2011 [↗](#); Torres-Paz et al., 2019 [↗](#)).

While the comparative anatomy of fish brains is reasonably well documented, behavioral studies have mainly focused on a few model species, primarily zebrafish (Kalueff et al., 2013 [↗](#)). The behavioral correlates of sensory systems diversity and evolution in fishes are poorly described and understood, which hampers the interpretation of cross-species comparisons. For example, the size and complexity of fish olfactory organs is highly variable and correlates with the richness of odorant receptors repertoire, defining a morpho-genomic space in which olfactory specialists and non-specialists species distribute (Burguera et al., 2023 [↗](#); Policarpo et al., 2022 [↗](#)). On the lower end, sunfish or pipefish have a small and flat olfactory epithelium and possess ~30 odorant receptor genes. On the higher end, polypteriforms have large and complex olfactory rosettes with up to 300 lamellae and possess ~1300 odorant receptor genes. Yet, the differences in olfactory behaviors, sensitivity and preferences between these species are completely unknown.

To start addressing the question of the evolution of olfactory sensory-driven behaviors in an amenable laboratory fish model, we used the blind cave and the river dwelling morphs of the Mexican tetra, the characiform *Astyanax mexicanus*. The species has become an established model for evolutionary biology at large, including evolution of behaviors (Duboué and Keene, 2016; Hinaux et al., 2016b; Kowalko, 2020; Yoshizawa, 2016). Cavefish embryos and larvae have larger olfactory pits than surface fish and their neuronal composition is changed (Blin et al., 2018; Hinaux et al., 2016a), while adults of the two forms have a similar olfactory rosette with 20-25 lamellae (Schemmel, 1967). The recent divergence between the two forms probably did not allow for a substantial evolution of their olfactory receptor gene repertoire, which is almost identical in the two eco-morphotypes (245 genes in cavefish, 233 genes in surface fish, from recent genome assemblies) (Policarpo et al., 2022). Yet, cavefish larvae tested in groups display outstanding olfactory detection capacities, as they are attracted to extremely low concentration of the amino acid alanine (10^{-10} M) while surface fish larvae have a more “classical” threshold (10^{-5} M) (Hinaux et al., 2016a) which approximates levels of free amino acids found in natural waters (Hara, 1994). In wild natural caves too, groups of cavefish adults respond to low concentrations of odorants (Blin et al., 2020). We hypothesized that the evolution of olfactory system development and olfactory skills in blind cavefish is an adaptive trait that, together with other constructive sensory changes in mechano-sensory and gustatory systems (Varatharasan et al., 2009; Yamamoto et al., 2009; Yoshizawa and Jeffery, 2011; Yoshizawa et al., 2014), may have contributed to their adaptation and survival in the extreme environment of their subterranean habitat. Focusing on olfaction, here we developed a sensitive, high-throughput behavioral assay and a pipeline for analysis to compare the types of behavioral responses elicited by food-related, amino acid odors in cavefish and surface fish, at population and individual levels. We determined their olfactory preferences and sensitivity thresholds for six amino acids, and we analyzed their behavioral responses in detail. We also described olfactory-driven behaviors in second-generation (F2) hybrids resulting from crosses between surface and cave morphs, as an attempt to understand the genetic component of the “olfaction trait” in *Astyanax* morphs. We discovered that the behaviors triggered by odorant stimulation has markedly evolved in cavefish.

Results

A high-throughput, sensitive behavioral assay to compare individual’s behaviors in *Astyanax* morphotypes

Our goal was to compare 1) olfactory discrimination capacities, 2) odor preferences and 3) behavioral responses to olfactory stimuli in Pachón cavefish (CF), surface fish (SF) and F2 hybrids (F2) individuals. As the three types of fish are markedly different in terms of basal swimming activity and patterns, we first sought to characterize the diversity of these behaviors in order to be able to interpret accurately fish responses to odors.

Six weeks old fish were habituated either 1h or 24h in their individual test box and they were first recorded for 30 minutes in the dark, without any stimulus (Figure 1; see Methods). We observed and categorized several typical and distinctive “baseline” swimming behaviors for individual fish (Figure 2A): random/haphazard swim (R), wall following (WF; defined as the fish continuously following the 4 sides of the box and turning around it, in a clockwise or counterclockwise fashion), large or small circles (C), thigmotaxis (T, along the X- or the Y- axis of the box; defined as the fish swimming back and forth along one of the 4 sides of the box), or combinations thereof. The distribution of these different types of swimming patterns was significantly different in Pachón CF, SF and F2 types of fish (Figure 2B; Fisher’s exact test; see also Methods and Suppl. Fig8A for another representation using Correspondence Analysis). A majority of SF swam in a random pattern (blue shades), while a majority of CF performed wall following (red/brown shades) and F2 fish showed more diversified swimming patterns. Importantly, the distribution of these baseline-swimming patterns was affected as a function of the habituation time, in all three types of fish

(**Figure 2B** [↗](#); $p=0.044$ for SF, $p=0.0005$ for CF and $p=0.0005$ for F2, Fisher's exact tests). Moreover, thigmotaxis behavior, which we have previously shown to represent stress behavior (Pierre et al., 2020 [↗](#)), was frequent in SF after 1h habituation (70% of individuals) but reduced after 24h acclimation (37%). In the same line, the swimming speed was different for all fish types when computed after 1h or after 24h of habituation (**Figure 2C** [↗](#)). This suggested that natural, unperturbed behavior is better observed after long, 24h habituation, which we applied thereafter.

As described above at population level, cavefish, surface fish and F2 displayed overall markedly different basal swimming patterns (**Figure 2B** [↗](#); see also Methods and **Suppl. Fig8A** [↗](#) for another representation using Correspondence Analysis). Yet, at individual level, there was a substantial degree of variability across individuals within a morphotype. That is, even though most SF tended to swim randomly (75% after 24h habituation; blue shades on graphs) and most CF performed wall following (50%; red/brown shades), these behaviors were not exclusive as some SF individuals displayed thigmotaxis behavior or some CF individuals swam in circles. Moreover and importantly, the type of swim pattern of each individual fish was remarkably stable and reproducible along several days of recordings, and swimming kinetics (position in the box, swimming speed, number of round trips in X and Y) showed no or very little variation (**Figure 2D** [↗](#) and **Suppl. Fig1AA'** [↗](#)). Together, these observations strengthened the importance of 1) using long habituation times, and 2) studying individual behaviors. They further highlighted an overlooked aspect of fish behavioral analyses: fish may have a “personality” that we sought to take into account when comparing behavioral responses to odors below.

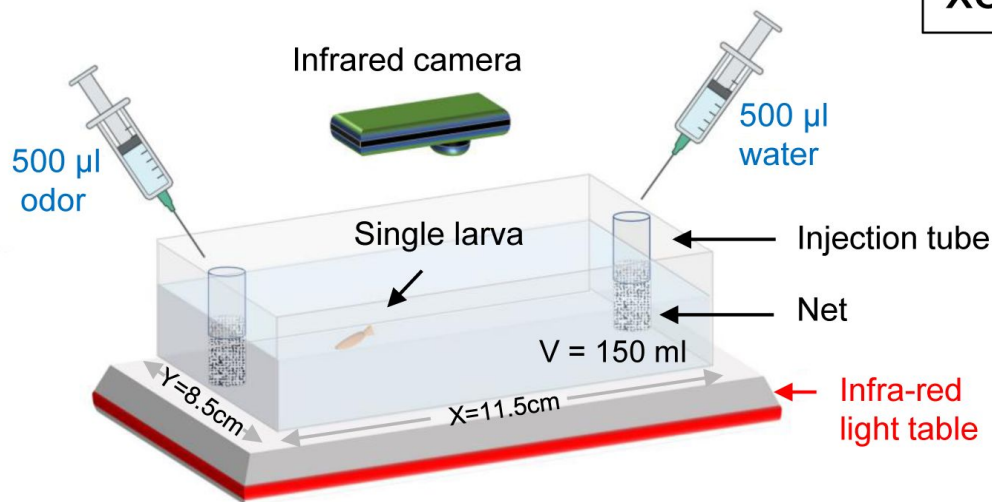
Finally, Pachón CF, SF and F2 hybrids possess markedly different sensory apparatuses and capacities, not only for chemo- but also for mechano-sensation (Lunsford et al., 2022 [↗](#); Yoshizawa et al., 2010 [↗](#)). Therefore, to test behavioral responses specific to olfactory stimuli, we wanted to find a way to deliver odors that would be vibration-less. We choose to inject odorant solutions inside tubes attached at the two extremities of the test box, designed to stop the wave and flow that would otherwise arise at the surface of the water upon injection (**Figure 1** [↗](#)). The efficacy of the device was demonstrated by the lack of perturbation in swimming behavior, neither qualitative (i.e., behavioral response and swimming pattern; **Figure 3AB** [↗](#)) nor quantitative (i.e., swimming kinetics; **Figure 3C** [↗](#)) for the 3 types of fish when water/control injections were performed on one side of the test box (or two sides; **Suppl. Fig1BC** [↗](#)). This suggested that our way to deliver odorant solutions did not generate vibratory perturbations. Moreover, when no injection was made, individual fish behaviors were stable over one hour of test (**Suppl. Fig1D** [↗](#)), allowing to observe accurately the effects of an olfactory stimulus if applied after 30 minutes. Finally, all recordings were performed in the dark under infra-red lights to neutralize the visual modality in sighted fish, a procedure which did not affect SF behavior (**Suppl. Fig1E** [↗](#)). Hence, the behavioral responses to odors recorded thereafter in Pachón CF, SF and F2 are driven by olfaction and can be compared (see also Hinaux et al., 2016a [↗](#)).

Behavioral responses to alanine in *Astyanax* morphotypes

Amino acids are food-related odorant cues for fish. Next, we characterized individual behavioral responses elicited by different concentrations of alanine, a potent aliphatic amino acid cue for most fish species including *Astyanax*.

Injection of alanine 10^{-2} M, 10^{-3} M, 10^{-4} M or 10^{-5} M (hence 10^{-4} M, 10^{-5} M, 10^{-6} M or 10^{-7} M final concentration in the odorant third of the testing box, respectively) induced a strong behavioral response in cavefish (**Figure 4A** [↗](#), 10^{-4} M injection shown; **Suppl. Fig2A** [↗](#) for other examples). Upon stimulus, CF decreased the numbers of back-and-forth swims along the X- and the Y-axes of the box (**Figure 4A** [↗](#), top left graph and **Figure 4FG** [↗](#)), and they shifted and restricted their swimming activity towards the odorant side of the arena, close to the odor delivery tube (**Figure 4A** [↗](#), bottom left graph, and **Figure 4E** [↗](#)). A tendency to decreasing swimming speed was observed but was significant only for 10^{-3} M injections (**Figure 4D** [↗](#)). Alanine injection also

A Setup



B Experimental timeline

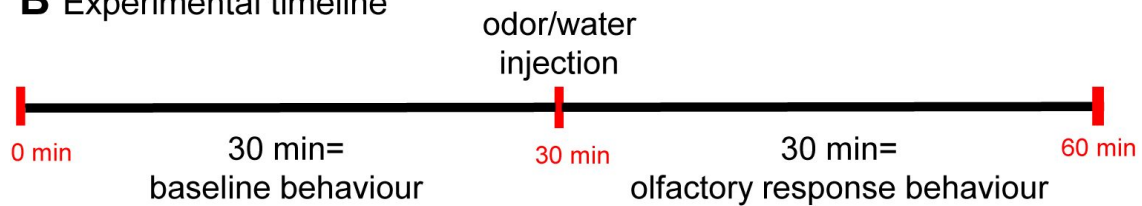


Figure 1

experimental setup for testing behavioral responses to olfactory stimulation in individual, 6 weeks old, *Astyanax mexicanus* larvae.

A, Each fish is placed individually in 150ml water in a rectangular testbox placed on an infrared light emitting table. Control or odorant solution are delivered at the two extremities of the box, inside tubes covered with a net. Medium to high throughput behavioral testing is achieved by parallel recording of 32 testboxes placed under 8 infrared recording cameras (4 testboxes per camera).

B, After one hour or 24 hours habituation, the test consists of one hour IR video recording. The first 30 minutes provide the control/baseline behavior of individual fish. Odorant stimulation is given at 30 minutes, and the behavioral responses are recorded for 30 more minutes.

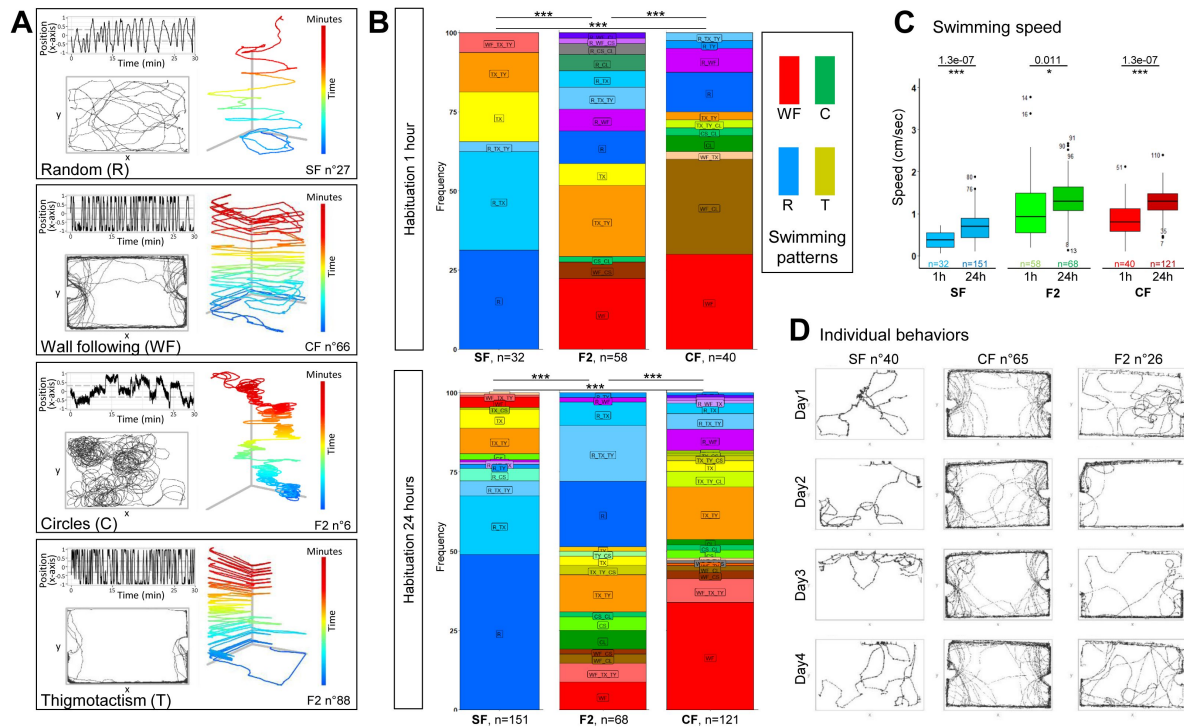


Figure 2

basal swimming behaviors of 6 week-old CF, SF and F2.

A, Examples of typical and distinctive basal swimming patterns exhibited by *Astyanax mexicanus* morphotypes. The behavioral patterns (Random, Wall Following, Circles, Thigmotactism) and the individual fish shown are indicated. Each pattern is best described by the combination of 3 types of representations. (1) the top left track shows the displacement of the fish along the x-axis of the box, representing back-and-forth swims along its long dimension. (2) the bottom left track is the top view, 2D representation of the trajectory. (3) the right track is the 2D plus time (color-coded) representation and helps, for example, to discern between wall following and thigmotactism.

B, Distribution of basal swimming pattern displayed by SF, F2, and CF after 1h or after 24h of habituation. Numbers of fish tested are indicated. The elementary color code is indicated (R in blue, WF in red, C in green, T in yellow). Exact swimming patterns and combinations are given on the colored plots (Suppl. Fig7 for the full color code). Fisher's exact tests for statistical comparisons between groups. Of note, in this and subsequent figures, the swimming pattern color code does not relate whatsoever with the time color code used in the 2D plus time representation of swimming tracks such as in panel A.

C, Boxplots showing the swimming speed in SF (blue), F2 (green) and CF (red), after 1h or 24h of habituation. Values are mean speed calculated over a period of 15 minutes. Numbers of fish tested are indicated. Mann-Whitney tests p-values are shown.

D, Examples of the stability of the basal swimming pattern over 4 experimental days in 3 individuals, with 24h of habituation time. One SF displaying Random pattern, one CF displaying Wall Following and one F2 displaying Thigmotactism + Random swim are shown.

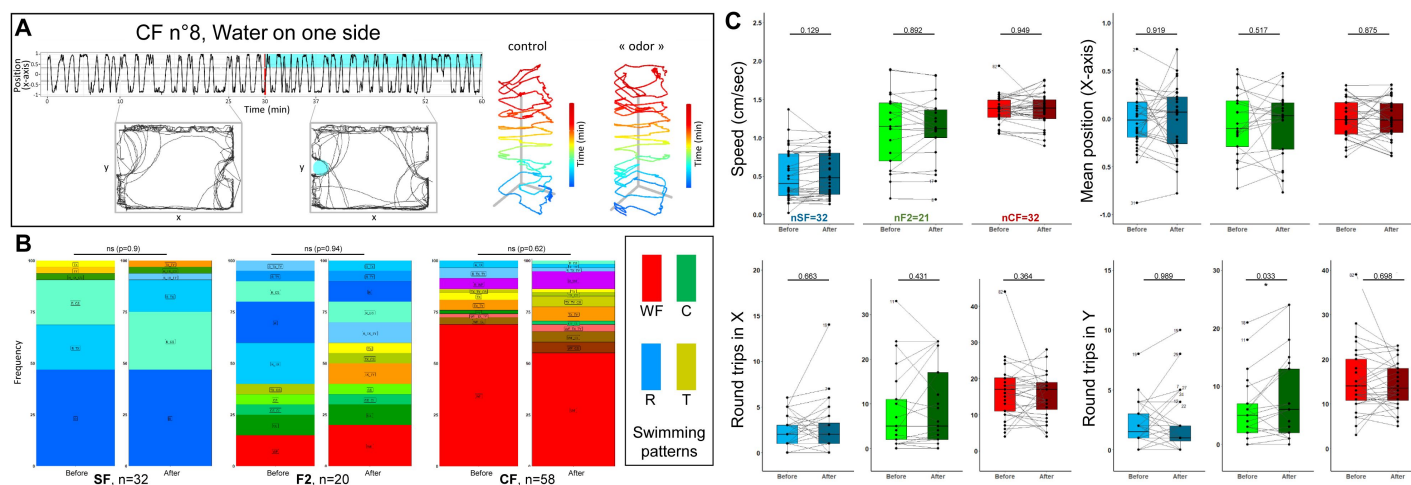


Figure 3

Controls on the olfactory setup and solution delivery method.

A, Lack of behavioral response in a representative CF individual after injection of water on one side of the testbox.

B, Lack of change in qualitative swimming pattern displayed by SF, CF and F2, before and after water injection on one side. Fisher's exact tests.

C, Lack of change in quantitative swimming kinetics in SF (blue), CF (red) and F2 (green), before (pale colors) and after (dark colors) water injection on one side. In this and the following graphs, boxplots show the swimming speed, the fish position along the X-axis of the testbox, and the numbers of back-and-forth swims along the X (length) and Y (width) axes of the box. All values are averaged over a 15 minutes period, either before or after the injection. A thin line links the value « before » and the value « after », for each individual. Numbers of fish tested are indicated on the "speed" boxplots. Numbers next to dots indicate the identity of outlier individuals.

markedly changed CF swimming patterns, as they completely switched from WF and T to R and C swimming modes (**Figure 4H** [↗](#) and **Suppl. Fig2D** [↗](#); Fisher's exact tests p values: 0.014, 0.0005, 0.0005 and 0.099 for 10^{-2} M, 10^{-3} M, 10^{-4} M or 10^{-5} M alanine injections, respectively; see also Methods and **Suppl. Fig8B** [↗](#) for another representation using Correspondence Analysis). In sum, CF individuals displayed robust behavioral responses and attraction to alanine. Of note, further decreasing the concentration of injected alanine to 10^{-6} M, 10^{-7} M, 10^{-8} , 10^{-9} M and 10^{-10} M had milder to no effects on cavefish. Swimming patterns remained globally unchanged (not shown), but swimming speed was decreased ($p = 0.005$ for 10^{-7} M; $n = 16$), numbers of round trips in Y were decreased ($p = 0.012$ for 10^{-7} M; $n = 16$), and numbers of round trips in X were decreased ($p = 0.017$ for 10^{-7} M; $p = 0.029$ for 10^{-8} M; $p = 0.003$ for 10^{-9} M; $n = 16$ each; data not shown). This suggested that in the present experimental setup, cavefish can detect very low concentrations of alanine, in agreement with previously published data ([Hinaux et al., 2016a](#) [↗](#)).

Conversely, surface fish responses were more subtle and seemed restricted to some individuals (**Figure 4B** [↗](#); **Suppl. Fig2B** [↗](#) for other examples). Notably, the numbers of back-and-forth swims along the X- and the Y-axes of the box were mostly unchanged and the position in the box did not vary for SF upon alanine injection (**Figure 4B** [↗](#), top and bottom left graphs and **Figure 4E** [↗](#)-G). The swimming speed was unchanged (**Figure 4D** [↗](#)). Swimming patterns were globally unaffected upon alanine injection (**Figure 4H** [↗](#) and **Suppl. Fig2D** [↗](#); Fisher's exact tests p values: 0.16, 0.54 and 0.80 for 10^{-2} M, 10^{-3} M or 10^{-4} M alanine injections, respectively; **Suppl. Fig8B** [↗](#) for another representation using Correspondence Analysis). As shown on **Suppl. Fig2B** [↗](#), some rare SF individuals appeared to change their swimming mode upon alanine 10^{-2} M (high concentration) injection, suggesting that they did perceive and react to the odorant stimulus. However, these rare individual responses were “diluted” at population level by the pooling of all fish in the distribution graphs. In summary, SF behavioral responses were modest and markedly different from CF.

Finally, F2 were tested with alanine 10^{-3} M, a concentration that elicits strong responses in CF but not in SF. Upon stimulus, F2's change in behaviors were similar to CF (**Figure 4C** [↗](#); **Suppl. Fig2C** [↗](#) for other examples): they decreased back-and-forth swimming activity, decreased swimming speed, and swam close to the odor source (**Figure 4D-G** [↗](#)). They also shifted their swimming patterns (**Suppl. Fig2D** [↗](#); Fisher's exact test $p = 0.0065$), including a loss of wall following mode that was reminiscent of the trend observed in CF. As an illustration, the individual shown in **Figure 4C** [↗](#) displays a striking change from a “large circles” to a “random” swimming pattern accompanied with variations in swimming kinematics.

Overall, these data suggested that, compared to SF, CF detection threshold and behavioral responses to the amino acid alanine have significantly evolved.

Behavioral responses to serine and cysteine in *Astyanax* morphotypes

We next systematically tested behavioral responses to injections of 10^{-2} M, 10^{-3} M or 10^{-4} M of serine (polar amino acid, hydroxyl group) and cysteine (non-polar, sulfur containing), two other potent amino acid olfactory cues for fish (**Figure 5** [↗](#) and **Suppl. Fig3** [↗](#)).

Serine elicited behavioral responses similar to alanine: CF (as well as F2 hybrids, **Suppl. Fig3** [↗](#)) moved towards the odor source, whereas SF did not (**Figure 5AB** [↗](#) and **D** [↗](#)). Swimming speed was unchanged in all fish types (**Figure 5C** [↗](#)), but a decrease or an increase of back-and-forth swimming activity was observed in CF and SF, respectively (**Figure 5E** [↗](#)). At population level, significant changes in swimming patterns were observed for the three fish types upon 10^{-2} M serine (**Figure 5G** [↗](#); Fisher's exact p-values 0.0005, 0.001 and 0.0085 for SF, CF and F2, respectively; **Suppl. Fig8B** [↗](#) for Correspondence Analysis). Of note, and contrarily to alanine, the lower concentrations of serine tested (10^{-4} M and 10^{-3} M in syringe; 10^{-6} M and 10^{-5} M in box,

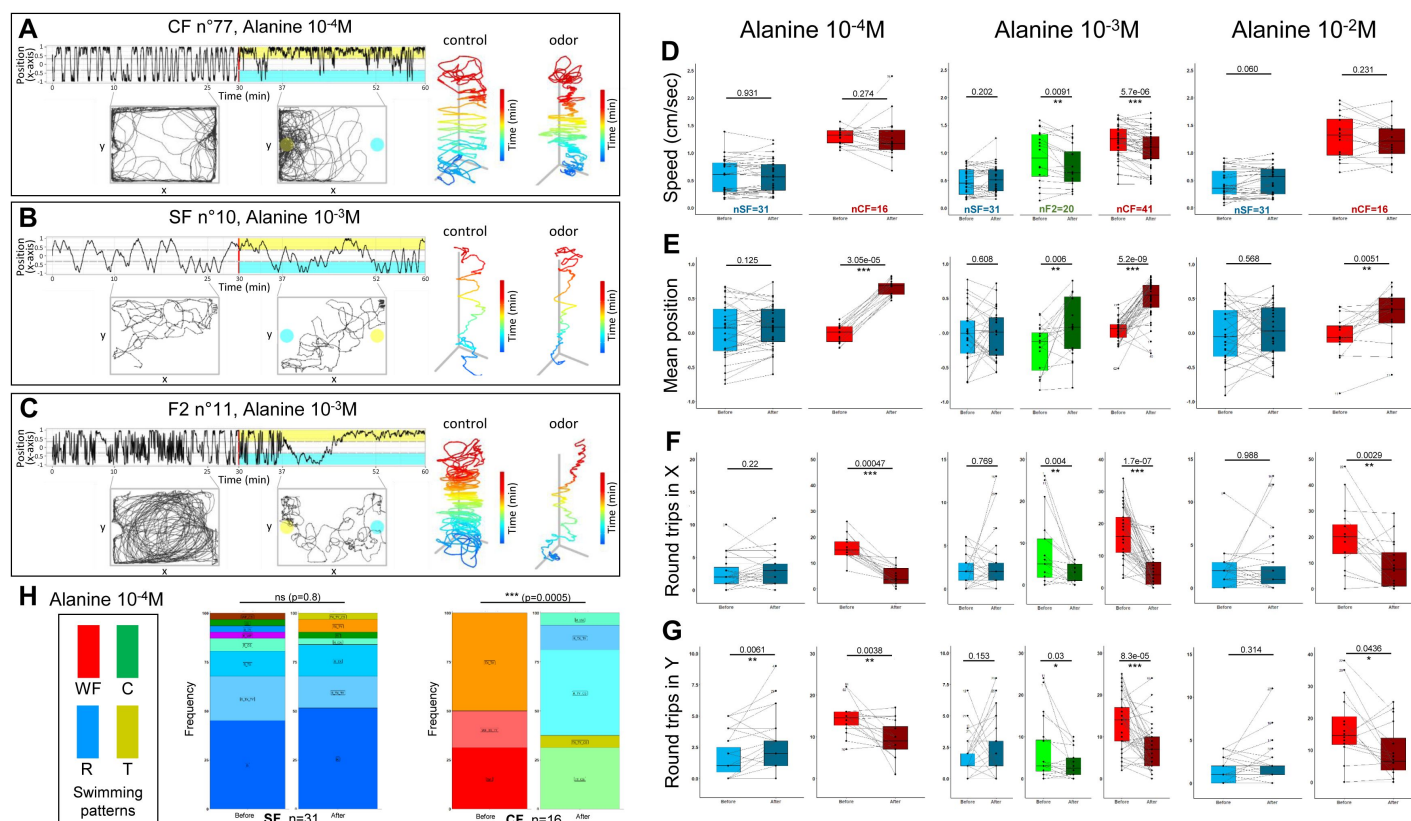


Figure 4

Behavioral responses to alanine of 6 week-old CF, SF and F2.

A,B,C: Representative individual responses of CF, SF and F2 after injection of alanine at the indicated concentrations. In the two left graphs of each panel, the blue color indicates the water/control injection side and the yellow color indicates the alanine injection side.

D,E,F,G: Box plots showing swimming speed (D), mean position along the X-axis (E) and the number of back and forth trips in X and Y (F,G) in SF (blue), CF (red) and F2 (green), before (lighter color) or after (darker color) the injection of alanine at the indicated concentration. Values are calculated over a 15 minutes period. Black lines link the « before odor » and « after odor » value for each individual. Numbers close to black dots indicate the identity of outlier individuals. P-values from paired Mann-Whitney tests are shown. The number of fish tested is indicated.

H: Change in swimming patterns exhibited by SF and CF after injection of alanine 10^{-4} M. Fisher's exact tests.

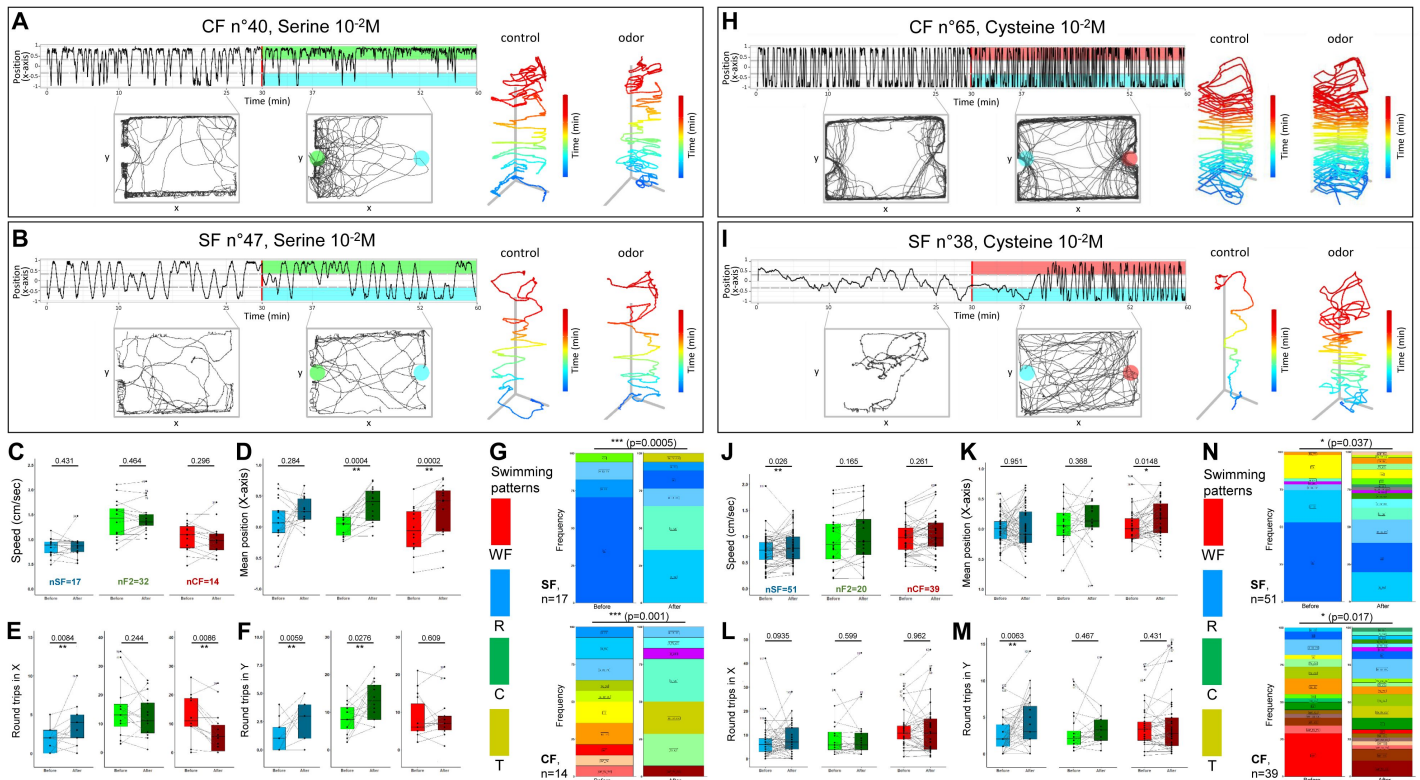


Figure 5

Behavioral responses to serine and cysteine $10^{-2}M$.

A,B and **H,I**, Representative individual responses of CF and SF after injection of serine (**A,B**) or cysteine (**H,I**) at high concentration ($10^{-2}M$). In the two left graphs, the blue color indicates the water/control injection side and the green (serine) or red (cysteine) color indicates the injection side.

C-F and **J-M**, Box plots showing swimming speed (**C,J**), mean position along the X-axis (**D,K**) and the number of back and forth trips in X and Y (**E,L** and **F,M**) in SF (blue), CF (red) and F2 (green), before (lighter color) or after (darker color) injection of serine $10^{-2}M$ (**C-F**) or cysteine $10^{-2}M$ (**J-M**). Values are calculated over a 15 minutes period. Black lines link the « before odor » and « after odor » value of each individual fish. Numbers close to black dots indicate the identity of outlier individuals. P-values from paired Mann-Whitney tests are shown. The number of fish tested is indicated.

G and **N**, Change in swimming patterns elicited after injection of $10^{-2}M$ serine (**G**) or cysteine (**N**) in SF and CF (F2 not shown). Fisher's exact tests.

respectively) were unable to trigger a robust behavioral change in cavefish (not shown, see Table1), suggesting that their detection threshold for serine is higher (not as good as) than for alanine. Regarding SF, responses were also observed only for 10^{-2} M serine injections.

Cysteine also produced significant effects (**Figure 5H-N** and **Suppl. Fig3**). Swimming speed and numbers of back-and-forth swims along the Y-axis increased only in SF and for the 3 concentrations tested (**Figure 5J** and **5M** for 10^{-2} M) (for 10^{-3} M: $p=0.01$ and $p=2e-05$, $n=32$; for 10^{-4} M: $p=2e-05$ and $p=0.00028$, $n=32$). CF were attracted to the odorant side for the 3 concentrations tested ($p=0.013$ to 0.025 ; $n=16$ to 38 each), whereas SF did not change position (**Figure 5K**) except for cysteine 10^{-3} M to which they seemed to be repelled ($p=0.0003$, $n=32$). The three fish types changed significantly their swimming patterns (**Figure 5N**; Fisher's exact p -values 0.03 , 0.017 and 0.04 for SF, CF and F2, respectively; **Suppl. Fig8B** for Correspondence Analysis). After injection of cysteine at the lower concentrations of 10^{-3} M or 10^{-4} M, CF behavioral responses were similar to those observed for alanine and serine, i.e., a decrease in back-and-forth swimming activity along the X-axis and a significant change in position towards the odor source, without changing swimming speed (not shown, see Table1). In sum, the different fish types show diverse responses to different concentrations of different amino acids.

Behavioral responses to lysine, histidine and leucine in *Astyanax* morphotypes

Finally, we choose to examine responses triggered by high concentrations (10^{-2} M) of three other, less studied amino acids: lysine, histidine (both polar and positively charged) and leucine (aliphatic, like alanine) (**Figure 6** and **Suppl. Fig4**).

For these three odors, CF responses were conspicuous and could include a shift of swim position towards the odor's source, a decrease of back-and-forth swimming activity with decrease in swimming speed, and significant changes in swim patterns at population level (except for leucine) (**Figure 6A-G** and **Suppl. Fig4**; **Suppl. Fig8B** for Correspondence Analysis). Regarding SF, changes were restricted to increases in back and forth swimming activity for histidine, without change in swimming speed or position in the box (**Figure 6A-D** and **Suppl. Fig4**). These three amino acids did not elicit changes in swimming pattern in SF at population level (**Figure 6E-G**). Finally, F2 also showed specific qualitative (swimming patterns) and quantitative responses to each of these three amino acids. Together, these data suggested that CF, SF and F2 detected and responded in their own and specific way to high concentrations of lysine, histidine and leucine.

Table 1 A provides a summary of the behavioral responses, averaged at population level, for the three fish types and the six amino acids tested. In the cases when significant behavioral responses were observed at population level, i.e., when responses are strongest and most stereotyped, different and opposed features emerged between CF and SF (and F2). As a “rule”, CF responses included a decrease in swimming activity (decrease in speed and back and forth swims along X- and Y-axes) together with an attraction to odor source and a change in swim pattern (**Table 1** A, red arrows). Conversely, SF responses mostly corresponded to an increase in swimming activity, with little or no attraction to odor and moderate change in swim pattern (**Table 1** A, blue arrows). In sum, a given sensory stimulus could trigger markedly different, morph-specific behavioral responses.

Behavioral responses to odors at individual level

Analyses as above performed at population level may mask or blur effects or phenotypes in the case when not all individuals respond in a stereotyped manner. As fish did express individual behavioral features in our experimental paradigm, we sought to perform further analyses at individual level, and to calculate individual response scores to the different odors. To do so, and to take into account the different components of the behavioral response, we summed the absolute

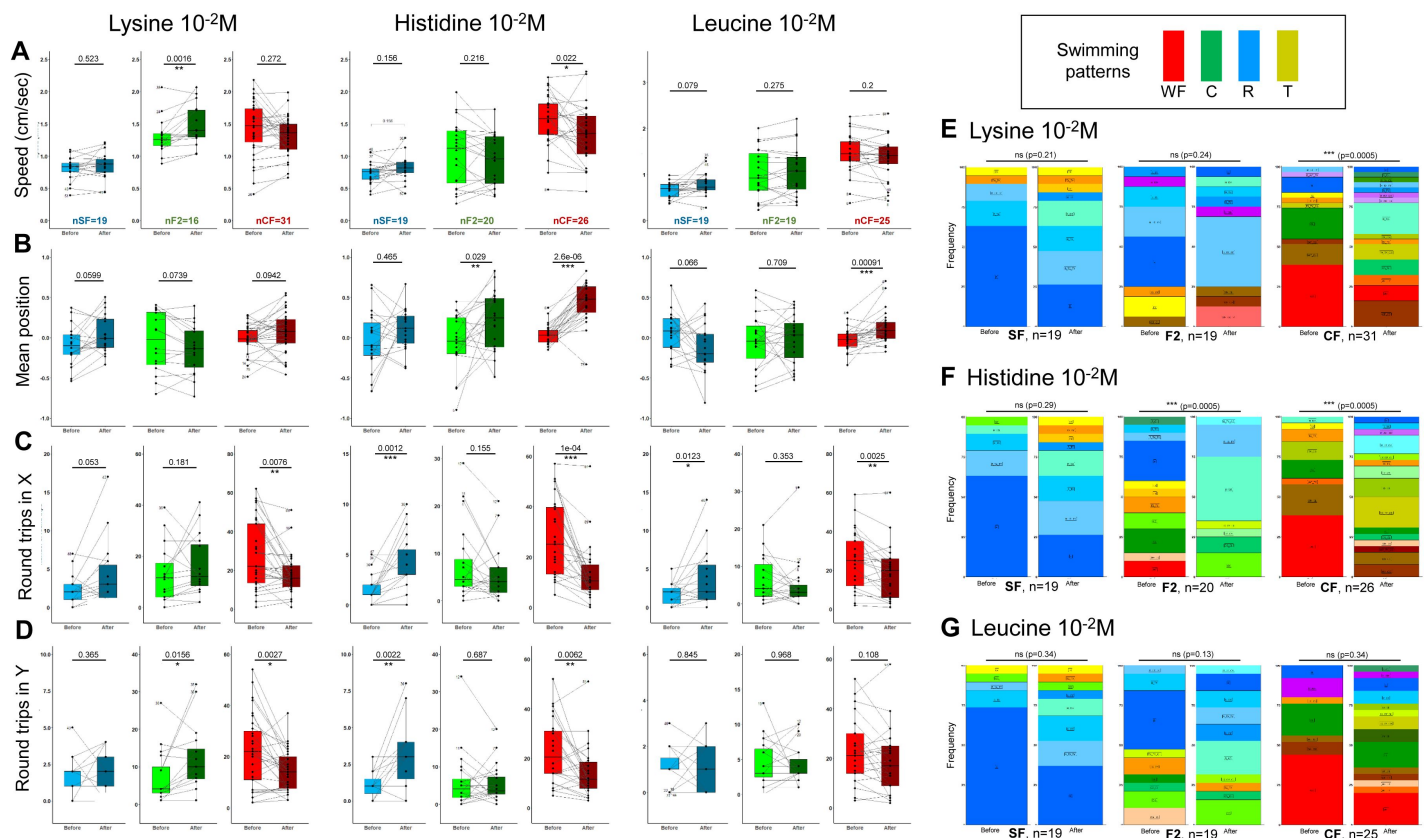


Figure 6

Behavioral responses to lysine, histidine and leucine $10^{-2}M$.

A-D, Box plots showing swimming speed (A), mean position along the X-axis (B) and the number of back and forth trips in X and Y (CD) in SF (blue), CF (red) and F2 (green), before (lighter color) or after (darker color) the injection of the indicated amino acid. Values are calculated over a 15 minutes period. Black lines link the « before odor » and « after odor » value of each individual fish. Numbers close to black dots indicate the identity of outlier individuals. P-values from paired Mann-Whitney tests are shown. The number of fish tested is indicated. See **Supplemental Figure 4** for examples of representative individual responses.

E-G, Change in swimming patterns elicited after injection of $10^{-2}M$ lysine (E) or histidine (F) or leucine (G) in SF, CF and F2. Fisher's exact tests.

		alanine 10 ⁻⁵ M	alanine 10 ⁻⁴ M	alanine 10 ⁻³ M	alanine 10 ⁻² M	serine 10 ⁻⁴ M	serine 10 ⁻³ M	serine 10 ⁻² M	cysteine 10 ⁻⁴ M	cysteine 10 ⁻³ M	cysteine 10 ⁻² M	lysine 10 ⁻² M	histidine 10 ⁻² M	leucine 10 ⁻² M				
A	population	Speed																
		SF	-	ns	ns	ns	-	↗	ns	↗	↗	↗	ns	ns	ns			
		F2	-	-	↘	-	-	ns	ns	-	-	ns	↗	ns	ns			
		CF	ns	ns	↘	ns	ns	ns	ns	ns	ns	ns	ns	↘	ns			
		Position																
		SF	-	ns	ns	ns	-	ns	ns	ns	↘	ns	ns	ns	ns			
		F2	-	-	↗	-	-	↗	↗	-	-	ns	ns	↗	ns			
		CF	↗	↗	↗	↗	ns	↗	↗	↗	↗	↗	ns	↗	↗			
		NbrtX																
		SF	-	ns	ns	ns	-	ns	↗	↗	↗	ns	ns	↗	↗			
		F2	-	-	↘	-	-	ns	ns	-	-	ns	ns	ns	ns			
		CF	ns	↘	↘	↘	ns	↘	↘	ns	↘	ns	↘	↘	↘			
		NbrtY																
		SF	-	↗	ns	ns	-	ns	↗	↗	↗	↗	ns	↗	ns			
		F2	-	-	↘	-	-	ns	↗	-	-	ns	↗	ns	ns			
CF	ns	↘	↘	↘	ns	ns	ns	ns	ns	ns	↘	↘	↘					
Patterns																		
SF	-	ns	ns	ns	-	ns	change	change	ns	change	ns	ns	ns					
F2	-	-	change	-	-	ns	change	-	-	change	ns	change	change					
CF	change	change	change	change	ns	ns	change	ns	change	change	change	change	change					
B	individual	Score > 1.5																
		SF	-	3.2%	19.3%	42%	-	25%	41.1%	18.7%	37.5%	51%	21%	42.1%	21%			
		F2	-	-	50%	-	-	31.3%	18.8%	-	-	55%	18.7%	55%	36.9%			
		CF	25%	81.3%	73.2%	56.3%	18.7%	6.2%	35.7%	31%	43.8%	39.5%	48.4%	69.2%	28%			

Table 1.

Summary of the results.

A, Population level summary. For each amino acid and each concentration tested, arrows indicate whether the considered parameter has changed (increased or decreased). ns indicates no significant change, - indicates the condition was not tested in the fish type (SF in blue, CF in red, F2 in green).

B, Individual level summary. The percentage of fish displaying an individual olfactory score superior to 1.5 is indicated.

values of indexes of speed, back and forth trips in X and Y, position and pattern changes (see Methods; **Figure 7A**). The threshold of score for which an individual fish was considered to respond significantly to a given stimulus was set at 1.5, from examination of the individual scores of fish in control conditions after perfusion of water (**Suppl. Fig5**). Visual inspection of the responses to amino acid odors for fish who had an individual score just above or just below this threshold confirmed that it was accurate.

For alanine 10^{-3} M, the representation of individual behavioral responses shows that CF responded in a very stereotyped manner (all lines following the same “curve”) whereas SF responses were more diverse (lines cross) (**Figure 7A**, 1st column). Moreover, 73% of CF but only 19% of SF had an individual score above threshold - a marked difference that was visible on the distribution of scores (**Figure 7AB**). F2 fish on the other end had a bimodal distribution of their olfactory scores for alanine 10^{-3} M (**Figure 7B**).

For other odors, the difference in olfactory scores of individual CF and SF was less obvious (**Figure 7B** and **Suppl. Fig5**), suggesting that taking into consideration individual's variations of swimming behavior can unmask responses that are not visible when averaged at population level. For example, in response to cysteine 10^{-2} M, SF (51% with score >1.5) and F2 (55% with score >1.5) had similar or slightly better scores than CF (39% with score >1.5) (**Figure 7AB**, 3rd column). Of note, these results had not shown up in **Table 1A** summarizing responses averaged at population level. Noteworthy, SF had mixed, diverse responses to all odors studied (lines crossing on all graphs). Conversely, CF showed non-homogenous responses to serine, histidine, leucine and lysine but highly stereotyped responses to both alanine and histidine (**Figure 7A** and **Suppl. Fig5**). A summary of individual response scores to the different odors is given in **Table 1B**.

Individual personality and individual behavioral response

The results presented above led us to test whether the “personality” of each fish, represented by their specific baseline swimming behavior, could influence their ability to respond to odors. To this end, we plotted individual olfactory scores as a function of individual baseline swimming patterns (R, WF, TX, TY, or C). We also examined possible correlations between basal swimming speed or round trip activity and individual olfactory score.

In SF, individual olfactory scores were similar whatever the basal swimming pattern of the fish (**Figure 8A**). However, for several odors, those fish exhibiting lower baseline swimming speed and less round trips in X had better olfactory scores (**Figure 8B** and **Suppl. Fig6A**; negative correlation). By contrast, in CF, fish exhibiting WF or C as baseline pattern had significantly better individual scores and those swimming randomly (R) had poor scores (**Figure 8A**) – but scores were not correlated whatsoever to baseline swimming speed (**Figure 8B**). Moreover, for alanine 10^{-3} M and histidine 10^{-2} M, the two odors for which more than 70% of CF showed a significant response (**Figure 7A**), a high number of round trips in X in baseline behavior was also associated to good olfactory scores (in agreement with WF being associated to significant individual response)(**Suppl. Fig6A**). Finally, F2 that exhibited a WF baseline pattern had better scores than others (like CF) (**Figure 8A**), and swimming speed was poorly correlated to their olfactory performances (not shown). Therefore, swimming patterns of CF and swimming speed of SF seem correlated to their behavioral response scores.

To further assess whether fish baseline swimming parameters are predictive of the behavioral output when presented with an odor, we next calculated how much of the odor response score variation can be explained by “personality”. We found that the PVE (Percentage of Variation Explained) on the individual olfactory score varies significantly with the baseline swimming pattern in CF (but not in SF) (**Figure 8C**) and with the swimming speed in SF (but not in CF) (**Figure 8D**). Altogether, these analyses strongly suggest that (1) the fish swimming personality

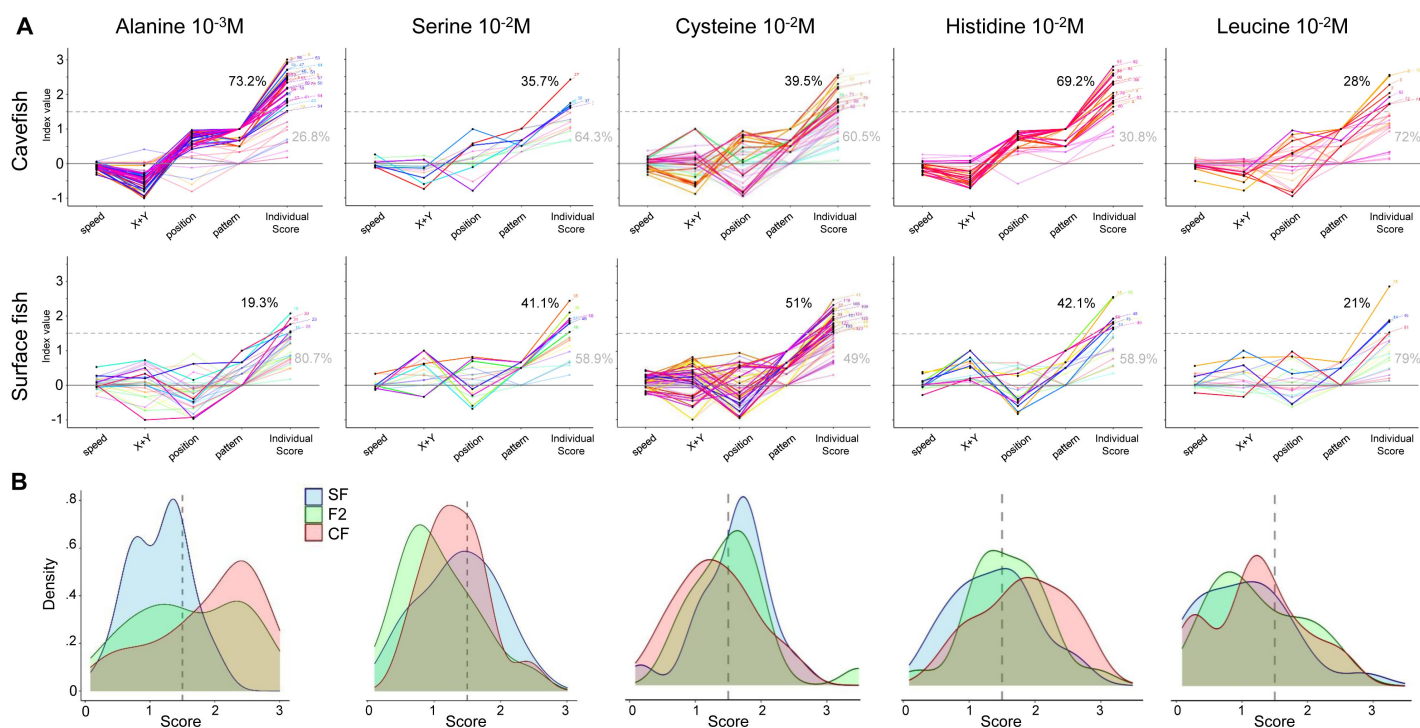


Figure 7

Individual olfactory scores of SF, CF and F2 for different odors.

A, Graphs representing index values (i.e., the variation between the 'before' and the 'after' odor condition; the value 0 corresponds to no change) for the 4 response parameters (speed, thigmotaxis in X and Y, position and swim pattern) used to calculate the total individual olfactory score for each fish (last column of points on each graph). Each fish is depicted by a colored line linking its 4 different indexes and its final individual olfactory score. The dotted line at value 1.5 indicates the score threshold above which a fish is considered as a responder. The percentages in black and grey indicate the proportion of responders/non-responders, respectively. The colored lines of responders are bright, those of non-responders are pale. Amino acids and concentrations are indicated. Top row: CF; bottom row: SF.

B, Distributions of individual olfactory scores of SF (blue), CF (red) and F2 (green) for different odors. The threshold score (1.5) is indicated by a dotted line.

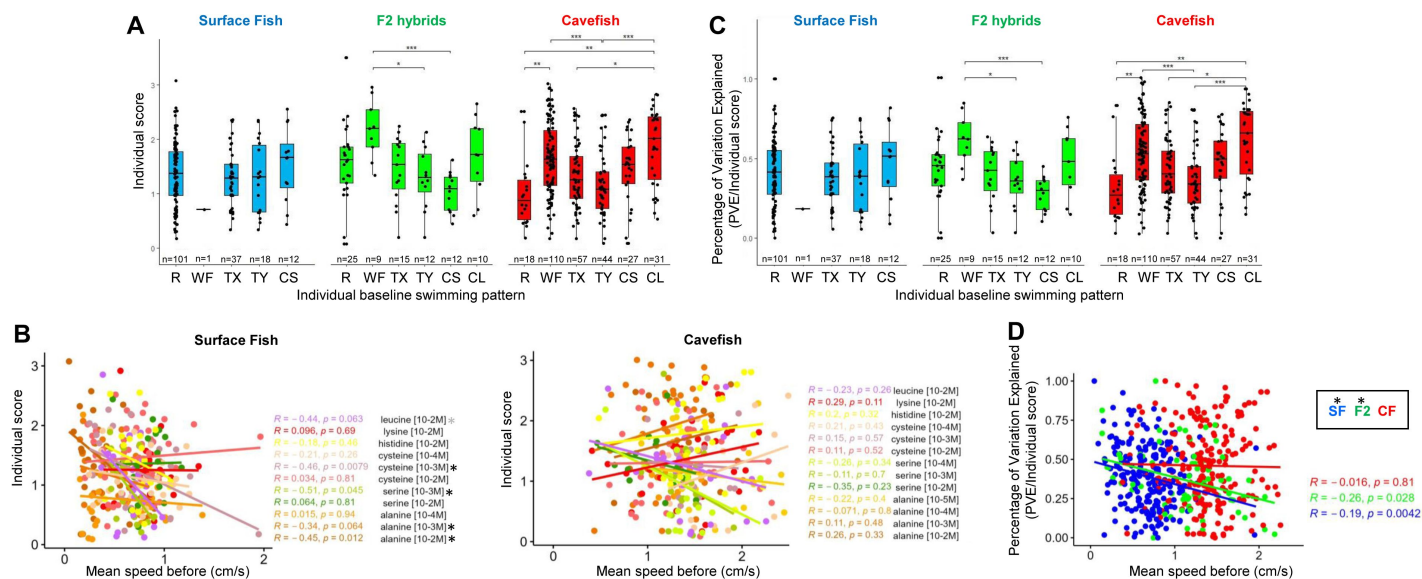


Figure 8

The individual olfactory score is related to individual fish swimming personality.

A, Box plots showing individual olfactory scores in SF (blue), F2 (green) and CF (red) as a function of their individual baseline swimming pattern. As stated in Methods, only amino acid conditions for which more than 40% of fish were responders (score >1.5) are pooled and plotted on this graph. Mann-Whitney two-tailed with Bonferroni post hoc tests were performed on each morphotype.

B, Regressions to explore the correlation between baseline swimming speed displayed by individual fish before odor injection and their individual olfactory score. Fish and amino acid type and concentrations are indicated. Linear correlations were calculated with Spearman's rank correlation test followed by Student test for the p values. Conditions for which significant correlation was found are labelled with an asterisk.

C, The PVE (Percentage of Variation Explained) on the individual score is plotted as a function of individual baseline behavior, to probe the predictability of the behavioral output as a function of "swimming pattern personality".

D, The PVE (Percentage of Variation Explained) on the individual score is plotted as a function of individual swimming speed, to probe the predictability of the behavioral output as a function of "locomotor activity personality". Linear correlations were calculated with Spearman's rank correlation test followed by Student test for the p values. Conditions for which significant correlation was found are labelled with an asterisk. SF (blue), F2 (green) and CF (red).

has an influence on its response to olfactory stimulation and can be predictive of the response, and (2) the personality parameter that is important to predict a good individual response is not the same in the two *Astyanax* morphs: in SF speed matters, while in CF swimming pattern matters.

Discussion

We have developed a high throughput, specific olfaction test, together with a pipeline of analysis allowing assessing qualitatively and quantitatively individual's and population's behaviors, in order to describe and compare responses of blind and sighted *Astyanax* to odorant cues. We discovered that cavefish, surface fish and F2 hybrids display different odor preferences and sensitivities and show individual, distinctive, diverse and specific responses to varying concentrations of the six amino acids tested. Using this novel setup where fish were tested in solo in a rectangular box and during one hour (as opposed to testing in groups in a U-shaped box and during 8 minutes in previous studies (Blin et al., 2020 [DOI](#); Blin et al., 2018 [DOI](#); Hinaux et al., 2016a [DOI](#)), we established that cavefish are *bona fide* “alanine specialists” and we analyzed in depth their behavioral responses.

A setup to probe olfaction in fish with markedly different sensory apparatuses and internal states

There is an inherent difficulty to compare sensory-driven behaviors in cave and surface morphs of *Astyanax*: all their sensory systems have evolved in one way or another. Cavefish have no eyes, but they possess enhanced mechanosensory lateral line and chemosensory gustatory and olfactory organs. To decipher behavioral responses driven by a single of these senses, one needs to control carefully the potential influence of the other sensory modalities. Here, we have recorded unimodal behavioral responses driven exclusively by olfaction by performing experiments in the dark to abrogate vision in sighted fish, and by designing a setup where the delivery of the olfactory stimulus is vibration-free. Moreover, we know from previous studies that gustatory taste buds do not participate in responses to amino acids at the concentrations used because the lesion of the olfactory epithelium abolishes the attraction to high concentrations of alanine, in both SF and CF (Hinaux et al., 2016a [DOI](#)).

Due to a mutation in their Monoamine Oxidase enzyme that interferes with the metabolism of brain monoamines (Elipot et al., 2014 [DOI](#)), Pachón cavefish have lower basal cortisol levels, hence lower basal anxiety than surface fish, when they are long habituated in their home tank (Pierre et al., 2020 [DOI](#)). However, the mutation confers cavefish with a much higher stressability after environmental change, such as the transfer in a novel tank (Pierre et al., 2020 [DOI](#)). Consequently, in order to record relevant behavioral responses to olfactory stimuli in unstressed fish, we used long acclimation times (72 hours) and we tested long habituation times, either 1 hour or 24 hours – as compared to 10 minutes as usually done in most fish studies including zebrafish or *Astyanax*. We concluded that 1 hour is too short for proper habituation, as the swimming speed and patterns were affected for both SF, CF and F2 as compared to 24 hours. Importantly, such long, 24 hours habituation periods are susceptible to reveal unperturbed and even novel behaviors: recently applied in a study of social behaviors in cavefish, 3 days of habituation allowed the analysis of behaviors in a familiar environment and could unmask social interactions in the so-called ‘asocial’ cavefish (Iwashita and Yoshizawa, 2021 [DOI](#)).

Neurophysiological and molecular considerations

In all vertebrates including fish, odorant molecules are recognized by olfactory receptors expressed at the surface of olfactory sensory neurons, which project onto the olfactory glomeruli in the olfactory bulb with the one receptor : one glomerulus rule (Axel, 1995 [DOI](#); Braubach et al., 2012 [DOI](#); Buck, 2000 [DOI](#); Kermen et al., 2013 [DOI](#); Koide et al., 2009 [DOI](#); Li et al., 2005 [DOI](#); Yoshihara,

2008). Moreover, parallel neural pathways in the olfactory circuitry process different types of odorants. In zebrafish, the perception of amino acids is mediated via Olfc/V2R receptors on microvillous sensory neurons that innervate lateral and ventro-medial glomeruli in the bulb. From the periphery to the brain, *Astyanax* surface and cave morphs display some variations in this amino acid signal processing circuitry. SF have 43 and CF have 41 V2R/Olfc receptor genes in their genomes (Policarpo et al., 2022), a minor difference that is unlikely to underlie their differences in olfactory capacities and preferences. CF larvae have higher proportions of microvillous neurons than SF, which together with the larger size of their olfactory epithelium and olfactory bulbs may influence their olfactory sensibility (Blin et al., 2018). *Astyanax* glomerular organization is unknown.

Interestingly, here we have found that CF are strongly attracted and respond to alanine and histidine, two amino acids which, albeit probably not recognized by the same receptor(s), are processed in the same or very close glomeruli in zebrafish larval olfactory bulbs (Li et al., 2005). SF on the other hand seem to show a preference for cysteine (sulfur), the amino acid that is the most potent to evoke electrical responses in the olfactory bulbs of sea breams (Hubbard et al., 2011) or hammerhead sharks (Tricas et al., 2009), and that elicits a strong aversive behavioral response in larval zebrafish (Vitebsky et al., 2005). Therefore, odor preferences have evolved between cavefish and surface fish, as well as between zebrafish and *Astyanax*. In the same line, previously we had reported that chondroitin is a strong attractant for *Astyanax* (Blin et al., 2020), whereas it induces freezing and fear behavior in zebrafish (Mathuru et al., 2012). Such significant variations in odor preferences or value may be adaptive and relate to the differences in the environmental and ecological conditions in which these different animals live. Of note, we have not found an odor that would be repulsive for *Astyanax* so far, and this may relate to their opportunist, omnivorous and detritivore regime (Espinasa et al., 2017; Marandel et al., 2020). However, the reason why Pachón cavefish have become “alanine specialists” remains a mystery and prompts analysis of the chemical ecology of their natural habitat. Alternatively, specialization for alanine may not need to be specific for an olfactory cue present only, or frequently, or in high amounts in caves. Bat guano for example, which is probably the main source of food in the Pachón cave, must contain and release many amino acids. Enhanced recognition of only one of them - in the present case alanine but evolution may have randomly acted for enhanced recognition of another amino acid - should suffice to confer cavefish with augmented sensitivity to their main source of nutriment.

Cavefish have also evolved regarding olfactory sensitivity. They are able to detect very low concentrations of alanine (aliphatic), hence they have a lower detection threshold than SF. Injection of low concentration alanine $10^{-4}\text{M}/10^{-5}\text{M}$ (thus $10^{-6}\text{M}/10^{-7}\text{M}$ in the odorant third of the arena) elicits strong behavioral responses in CF, whereas even higher concentrations of 10^{-3}M and 10^{-2}M evoke modest responses in SF (zero response at population level, 19% responders at individual level with alanine 10^{-3}M). We have performed dose-response experiments for 3 out of the 6 amino acids tested and for both morphs. With these, CF also appear capable of detecting low concentrations of cysteine (sulfur). Moreover, they seem to detect better histidine 10^{-2}M (polar) as well (69.2% of CF responders, versus 42.1% of SF), and CF but not SF detected lysine (polar) at the relatively high concentration of 10^{-2}M . From these observations, we can predict that the difference between SF and CF probably does not lie in the molecular evolution of their olfactory receptor repertoire, because it is unlikely that all the receptors recognizing diverse types of amino acids have evolved all at once. Rather, we can hypothesize that evolution occurred at the level of the general regulation of odorant receptor genes expression, or at the level of olfactory processing and computation in the bulbs (Friedrich et al., 2009), or in higher order brain regions.

A variety of behavioral responses to olfactory stimulation

Our study suggests that behavioral responses to a unimodal olfactory sensory stimulus are complex and have evolved in cavefish (summary on Table 1). When they detect the odor stimulus, cavefish globally decrease their scanning activity, which might help them to compute

and swim up towards the higher values of concentration in the odor plume. This is associated to a change in swimming pattern whereby WF and T are eliminated at the expense of R and C, presumably to facilitate searching. This hypothesis is further supported by their systematic change of position in the box, meaning that they locate efficiently the odor source. Strikingly, surface fish display opposite behavioral responses after odor detection: they rather increase locomotor activity, which corresponds to intense foraging but does not seem optimal to find the odor source, which is confirmed by their lack of change in position of the box (i.e., they may not locate efficiently the odor source). This is also consistent with the fact that at individual level the best SF responders are slow swimmers. Such poor ability to find the odor source may result from testing in the dark. Indeed, SF behaviors are mostly visually driven (but see [Simon et al., 2019](#)) and they might need multimodal visual + olfactory integration to find food efficiently. These interpretations are consistent with early work showing that, when competing in the dark and with limited amounts of food, SF starve and CF thrive ([Hüppop, 1987](#)). In sum, CF foraging strategy has evolved in response to the serious challenge of finding food in the dark. Future experiments including functional imaging of brain activity in live animals may reveal the changes in olfactory-driven motor circuits that allowed the evolution of behavioral outputs in cavefish.

Finally, at population level, F2 fish shared some behavioral response traits with both parental morphs, and they were often closer to CF. At individual level their responses were also variable and non-stereotyped (including for alanine 10^{-3} M and histidine 10^{-2} M, the conditions for which CF showed highly stereotyped responses). Olfactory scores behave as a quantitative trait. The tools we have developed here will allow the future determination of the genetic underpinnings of the evolution of olfactory-driven cavefish behaviors and capabilities.

A “personality” for each fish?

Our recordings of several hundred ($n=489$ total) well-habituated, individual *Astyanax* larvae highlight an often overlooked aspect of fish behavioral analyses: fish may have a “personality” or “temperament”, which we characterized at the level of their baseline swimming patterns. This hypothesis is strongly supported by the consistency over time, one day after the other, of the favorite swimming pattern or combination of patterns and of the swimming kinetics expressed by individuals when freely swimming without stimulus (most fish were recorded on 4 days distributed over 2 weeks). To our knowledge, this is the first time individual temperament is taken into consideration in *Astyanax* behavioral studies.

In the context of ecology and evolution, the personality in non-human animals was proposed to be organized along five primary axes: sociability, boldness, aggressiveness, exploration and activity ([Réale et al., 2007](#)). In fish, studies applying an animal personality approach have focused to resolve variations in physiological and molecular parameters, suggesting a link between phenotype and genotype, or between behavior and transcriptome regulation eg ([Rey et al., 2021](#)). Here, we propose that the different temperaments expressed by individual larvae/juveniles could correspond to the genetic diversity present in the natural populations of *Astyanax*, a diversity that we have maintained intentionally along generations in captivity in our fish facility.

Further, we have assessed whether swimming temperaments could influence the way and the extent to which fish respond to an olfactory sensory stimulus. Strikingly, we discovered that baseline swimming patterns and swimming speed do influence fish olfactory responses. In addition, important personality traits that confer positive behavioral responses are not the same in SF and CF. For cavefish, WF and round trips in X (but not speed) are the key parameters. Wall following behavior has been reported in *Astyanax* and other blind cavefish species ([Chen et al., 2022](#); [Patton et al., 2010](#); [Sharma et al., 2009](#)) and was hypothesized to confer foraging advantages, although this had never been directly tested. Here, it seems to be the case for 6-week old juveniles in a rectangular box - but the link may be more elusive when considering a fish swimming in a natural, complex environment. We propose that WF and round trips in X are

together the expression of an intense exploratory behavior, which allows CF to better cover their swimming space and thus have a higher probability to detect odorant cues - even at very low concentrations because their detection capacities for some amino acids are excellent. By contrast, a slow swimming speed (but not swimming pattern) is the most critical personality parameter for SF. As SF olfactory concentration detection thresholds are higher than CF, one could imagine that they need more integration time inside the odorant plume to trigger a neuronal and a behavioral response. In any case and importantly, personality parameters that matter in CF and SF to confer good olfactory response scores are distinct. This suggests that the modulation of neuronal circuits underlying both the control of baseline behaviors and olfactory processing has evolved in cavefish.

Conclusion

Previous studies had shown that, when tested in groups, cavefish larvae as well as adults smell better than their surface fish conspecifics (Blin et al., 2020 [↗](#); Blin et al., 2018 [↗](#); Hinaux et al., 2016a [↗](#)). Here, we have established that it is also true for fish in solo, ruling out the possibility that when in group, one “good smeller” individual could drive others to respond through (unknown) communication mode. Using individual behavioral tests, we have discovered that both olfactory sensitivity, olfactory preferences, olfactory behavioral responses and key personality parameters have evolved in cavefish, conferring them with outstanding skills to forage in the darkness of caves.

Methods

Fish samples

Laboratory stocks of *A. mexicanus* surface fish (SF, origin: San Salomon spring, Texas, USA) and cavefish (CF, Pachón population) were obtained in 2004 from the Jeffery laboratory at the University of Maryland, College Park, MD, USA. Fish were maintained at 22°C with a 12h:12h light:dark cycle, fed twice daily with dry and live food. SF and CF brood stock were induced to spawn once every two weeks by thermal shock at 26°C. A F1 hybrid family was generated by crossing a SF female with a CF Pachón male. Second-generation hybrids (F2) were generated by crossing two F1 individuals. Embryos and larvae were reared at 24°C in water produced by our animal house's water production system. All fish used (CF, SF and F2) were exactly six weeks old on the first day of the experiment. SR's authorization for use of *Astyanax mexicanus* in research is 91-116. Animals were treated according to the French and European regulations for animal testing in research (authorization n°APAFIS#8604).

Experimental set-up

The test consisted of filming individual larvae for 30 minutes before olfactory stimulation and for a further 30 minutes afterwards (see Supplemental movies). The behavior laboratory is a soundproofed room maintained at a constant temperature of 24° C. To ensure that eyed fish (SFs and F2 with eyes) could not use the visual modality during the test (unlike CFs and some F2), all experiments (except one) were carried out in the dark. The experimental device was composed of 2 infrared (IR) tables (60x60cm each) over which 32 test boxes were placed. A custom-built wooden holder supported 8 IR cameras (DIGITNOW!), located 40 cm above the boxes. Each camera filmed 4 boxes simultaneously (**Figure 1** [↗](#)).

The individual test boxes (11.5x8.5x4.3 cm, multiroir #45103BOILAB03) were made of crystal-clear plastic. Odorant solutions (and/or water) were manually injected at the two opposite sides of the box using 1ml syringes. In order to stop the dispersion wave of injection, and to assure that we

always injected at the same right place, we added 2 injection guides in each box: we glued two plastic tubes (4 cm long and 1 cm in diameter) at the center of the two opposite short sides (Y), 0.2 cm from the bottom. In addition, a net (6.5x6.5 cm) was attached around each tube and reached the bottom of the box, to prevent fish from passing under the tube during the experiment, thus escaping from the subsequent tracking (**Figure 1** [↗](#)).

Behavior tests

Larvae were placed individually in a box containing 150 ml of water (clean rearing water), fed with micro-worms (three drops of concentrated *Panagrellus redivivus* solution placed in the center of the box) and subjected to a 12:12h day:night cycle for 72 hours acclimation. Then, the water in each box was changed, the boxes were placed on IR tables for 24 hours, and the larvae were fasted (=habituation). The day of testing, for reasons of reproducibility, all experiments started at 11 a.m. The light was switched off, the cameras were switched on and the experimenter left the room for a 30-minute baseline behavior recording period. At t=30 min, the experimenter entered the room and switched on the inactinic red lamps. Solutions were manually injected bilaterally (500 µl of odor/ 500 µl of water) into the injection guides of each box. The total time needed to perform injections of 32 boxes was 5 minutes. Then, the experimenter switched off the inactinic lamps and left the room for a 30-minute response behavior recording period. At the end of recording, the water was changed and each box was placed back exactly at the same location on IR table for 24 hours and not moved again until the following day of testing (the larvae were not fed). For the series of experiments with a habituation period of 1 hour, the water in the boxes was changed every morning, 1 hour before the start of the test. The typical timeline was as follows: [72 h acclimation and feeding], Week1 [habituation 24h, test day 1, habituation, test day 2, rest and feeding for three days], Week2 [habituation, test day 3, habituation, test day 4]. As the larvae were tested on several consecutive days, to ensure that the fish were not learning/associating one side with a scent cue, the odor injection side was reversed each day. The control experiments (unilateral or bilateral injection of 500 µl rearing water or no injection) were always carried out the day1 when the larvae were naive. For the “light vs dark” control experiment, SF were filmed 30 minutes in normal light and 30 minutes in the dark.

Odorant solutions

The odorant solutions were prepared by dilution of amino acids in clean rearing water (L- Alanine #A7627; L-Serine #S4500; L-Lysine #L5501; L-Histidine #H8000; L-Leucine #L8000 and L-Cysteine #168149; all from Sigma-Aldrich). A volume of 500µl (either water or odor) was injected with a 1ml syringe (Sigma-Aldrich #Z683531) fitted with a needle (0.8x50mm, 21G; B Braun #4665503). Depending of the test, concentration of solutions injected were 10^{-2} M or 10^{-3} M or 10^{-4} M. In a total water volume of 150ml, with a box virtually separated into three, the volume of water in the odor injection zone is 50 ml. By injecting 500 µl of solution into this zone, we expected to obtain a dilution factor of 100. For example, for a concentration of $[10^{-3}$ M] in the syringe, the concentration in the odor injection zone will therefore be $\sim[10^{-5}$ M]. All concentrations given in the article indicate the concentration inside the syringe.

When consecutive injections of an odor or different odors at different concentrations were tested on the same fish, they were performed from the least concentrated to the most concentrated.

Video editing and tracking

Videos were saved on SD card in AVI format (1208x720 px, 30 fps), then edited using Adobe Premiere Pro V14.0 to be calibrated at 1 hour long without the 5 minute-injection time. They were then exported into MPG2 format. The tracking software used was TheRealFishTracker V0.4.0 (<https://www.dgp.toronto.edu/~mccrae/projects/FishTracker/> [↗](#)), developed and freely available from the University of Toronto. In our hands, it was the only software able to detect properly transparent CF larvae. The parameters used were Confidence Threshold = 10; Mean Filter Size = 1;

signed Image = *dark object* and all other parameters were the default ones. The x/y-scales were drawn and given in cm. The output data file provided 30 X/Y coordinates (in pixels and in cm) per seconds for a 1-hour movie.

Graphical representations, quantifications and statistics

Statistical analyses were carried out with R-3.4.2 software (R Development Core Team, 2016 [↗](#)) using the stats and rstatix libraries, and all graphical representations were designed using the ggplot2, ggrepel, gghighlight and RGraphics libraries.

Quantitative parameters

From the output data set, pixel coordinates X_{pi} and Y_{pi} were first recoded using the box center as 0, X_{cmax} = 1 on the odor side and X_{cmin} = -1 on the water side. The same was applied to the Y coordinate with Y_{cmax} = 0.74 and Y_{cmin} = -0.74. Fish position in the box along the X-axis during test was calculated using X_c and Y_c at each time point (30/s). Speed between two time points (5-second time step) was calculated thanks to pythagoras's theorem using X/Y in cm. Round trip number is the total number of time the fish crosses the 0.5 and -0.5 position (along X-axis) divided by two. Idem for round trips in Y-axis. Quantitative parameter means were all calculated over a 15-minute period, from minute 10 to minute 25 for the 'before' period, and from minute 37 to minute 52 for the 'after' period. Means are represented by box plots showing the distribution of individual values of each fish (black dots), the median, the 25th- 75th percentile and outliers indicated by fish ID number. Paired Mann-Whitney two-tailed tests were performed to compare means of position, speed, or round trips before and after odor injection.

Non-paired Mann-Whitney two-tailed tests were performed to compare means of speed between 1h and 24h habituation periods represented in **Figure 2C** [↗](#) by box plots.

Swimming patterns

The swimming activity of the fish in the test box was systematically represented under 3 different forms, which allow grasping the details and different aspects of the behaviors (see **Figure 2A** [↗](#)). X_c / t coordinates were used with the geom_line() function for the 'Position (X-axis)' graph, (X_c,Y_c) / t coordinates were used with the geom_path() function for '2D' top view, and the gg3D package for '2D + time' view.

The determination of baseline swimming patterns and swimming patterns after odor injection was performed manually based on graphical representations such as in **Figure 2A** [↗](#) or **Figure 3A** [↗](#). Four distinctive baseline behaviors clearly emerged. (1) Random swim (R; defined as haphazard swimming with no clear pattern, covering entirely or partly the surface of the arena). (2) Wall following (WF; defined as the fish continuously following along the 4 sides of the box and turning around it, in a clockwise or counterclockwise fashion). (3) Large or small circles (C; self-explanatory). (4) Thigmotaxis (T, along the X- or the Y-axis of the box; defined as the fish swimming back and forth along one of the 4 sides of the box). On graphical representations of swimming pattern distributions, we used the following color code: R in blue, WF in red, C in green, T in yellow. Of note, many fish swam according to combination(s) of these four elementary swimming patterns (see descriptions in the legends of Supplemental figures showing many examples). To fully represent the diversity and the combinations of swimming patterns used by individual fish, we used an additional color code derived from the "basic" color code described above and where, for example R+WF is purple. The complete combinatorial color code is shown in **Suppl. Fig7** [↗](#). Of note, in all figures, the swimming pattern color code does not relate whatsoever with the time color code used in the 2D plus time representation of swimming tracks such as in **Figure 2A** [↗](#). Swimming patterns (i.e., qualitative variables; compositional dataset) in different

conditions were analyzed using Fisher's exact test comparing the quantity and distribution of different swimming patterns between SF/CF/F2 at day1, or over day1 to day4, or between 1h vs 24 h habituation period, or before vs after odor injection for each morphotype.

To reinforce our conclusions, swimming patterns in different conditions were also compared using CA (Correspondence Analysis), an appropriate method to analyze compositional data (Greenacre, 2021 [↗](#)). The results are plotted in **Suppl. Fig8** [↗](#). The distribution of swimming patterns in SF/CF/F2 in control experiments, or before vs after odor injection for each morphotype confirm the changes (or absence of changes) in behavioral patterns suggested by the colored bar plots in main Figures, with confidence ellipses overlapping or not overlapping, depending on cases. Of note, CA analysis cannot provide statistical support through calculation of p values.

Index and score

For quantitative swimming parameters: indexes of position, speed and number of round trips changes were calculated using the equation $(\text{mean after} - \text{mean before}) / (\text{mean after} + \text{mean before})$ commonly used in olfactory tests in fish (Choi et al., 2021 [↗](#); Koide et al., 2009 [↗](#); Wakisaka et al., 2017 [↗](#)). The index can vary from zero to 1. An index close to zero indicates that the values of the considered parameter before or after odor injection are close. The higher the absolute value of the index, the greater the difference. When the parameter increases after injection, the index is positive; when the parameter decreases after injection, the index is negative. The *absolute* value of each index will be used for score calculation below.

For qualitative swimming parameter: the index of pattern change was calculated as the number of new patterns observed after stimulation, divided by the total number of patterns observed after odor injection. An index equal to zero indicates no pattern change after stimulation. The higher the index is, the greater the difference of pattern is.

For each fish, the individual olfactory score for a given odor and concentration was the sum of the *absolute* values of the four indexes: $\text{individual score} = |\text{position index}| + |\text{speed index}| + |\text{round trips in X+Y index}| + |\text{pattern index}|$. On **Figure 7A** [↗](#) and **Suppl. Fig5** [↗](#), the 4 parameter's indexes and the total individual fish olfactory scores are represented by dots connected by a colored line for each individual. Those with a score > 1.5 are highlighted and considered to display a significant behavioral response to the stimulus. The 1.5 threshold value was arbitrarily determined after the examination of individual olfactory scores in control experiments (**Suppl. Fig5** [↗](#)) and the observation that the vast majority of fish have individual scores below this value in the absence of specific olfactory stimulus.

On **Figure 8A** [↗](#) and **8C** [↗](#), to assess the relationship between individual olfactory score and baseline swimming pattern, we pooled results of odors for which more than 40% of individuals have an olfactory score > 1.5 . So for SF, alanine [10^{-2}M] / cysteine [10^{-2}M] / histidine [10^{-2}M] / serine [10^{-2}M] have been used; for CF, alanine [10^{-2}M] / alanine [10^{-3}M] / alanine [10^{-4}M] / cysteine [10^{-2}M] / cysteine [10^{-3}M] / histidine [10^{-2}M] / lysine [10^{-2}M]; and for F2, alanine [10^{-3}M] / cysteine [10^{-2}M] / histidine [10^{-2}M] (see **Table 1B** [↗](#)). Mann-Whitney two-tailed with Bonferroni post hoc tests were performed on each morphotype.

Slopes in linear regressions show relationships between mean speed or number of round trips before injection and olfactory score. Linear correlations were calculated with Spearman's rank correlation test followed by student test for the p.values.

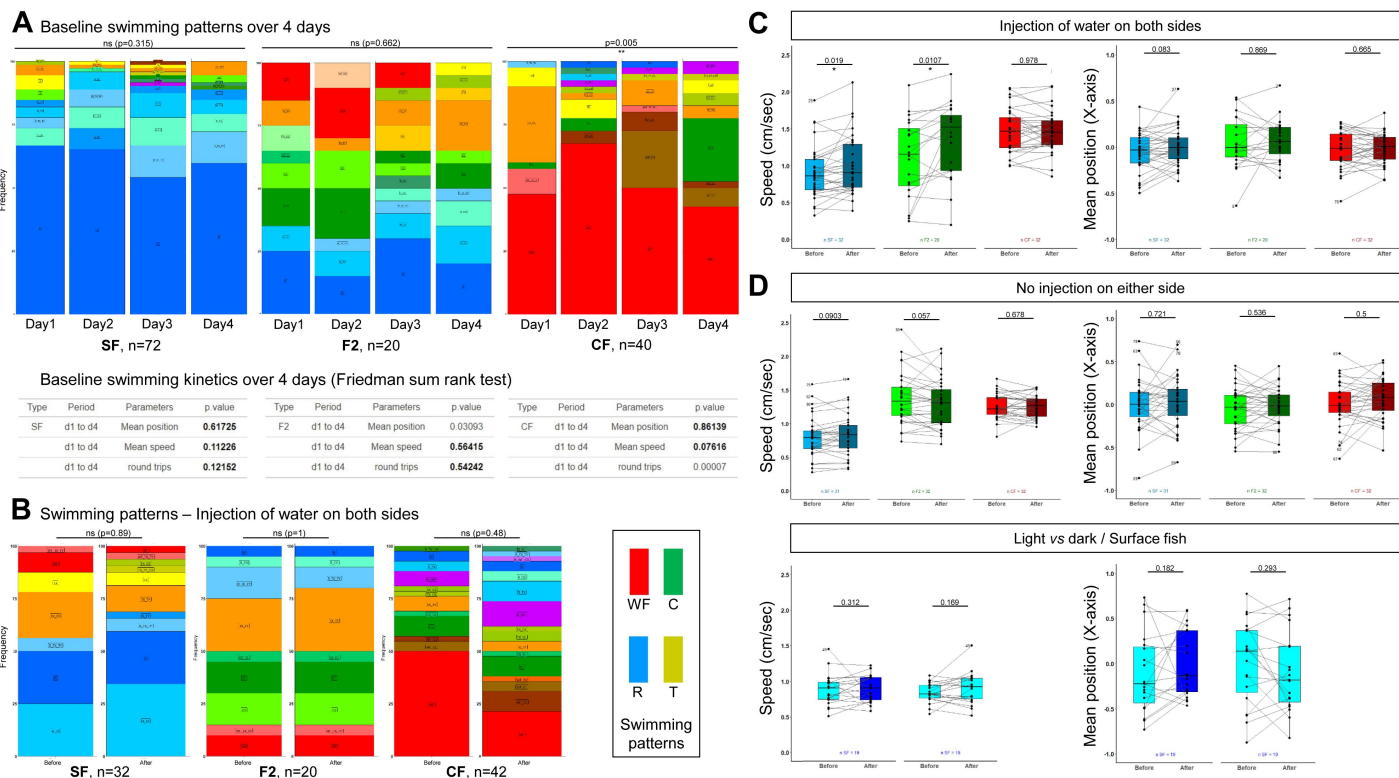
Supplementary information and data availability

The R script used to recode raw data into corrected coordinates, to calculate means and to draw 2D and 2D + time graphs is available on GitHub (<https://github.com/Maryline-Blin/Astyanax-swimming-pattern>).

F1 to F3 Files correspond to source data files (.txt) containing the raw data presented and analyzed in this paper. For each type, the files provide the ID numbers of the fish, condition (day1 to day4 or odor or control type), means of position, speed, X and Y round-trips and behavior for periods 10-25 minutes and 37-52 minutes, as well as the calculated individual olfactory score.

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We thank Camille Lejeune for her participation in the generation of the SF x CF F1 hybrid line, Maxime Policarpo for preliminary testing of tracking softwares and Rose Tatarsky for interesting discussions. Zootechnical care of our *Astyanax* fish facility was taken in charge by the TEFOR Unit. We are indebted to Joël Attia for invaluable help with statistical analyses during the revision of our manuscript. This work was supported by Equipe FRM (Fondation pour la Recherche Médicale) grant EQU202003010144 to SR.



Suppl Fig1 (with Figure 3).

Additional controls for experimental design.

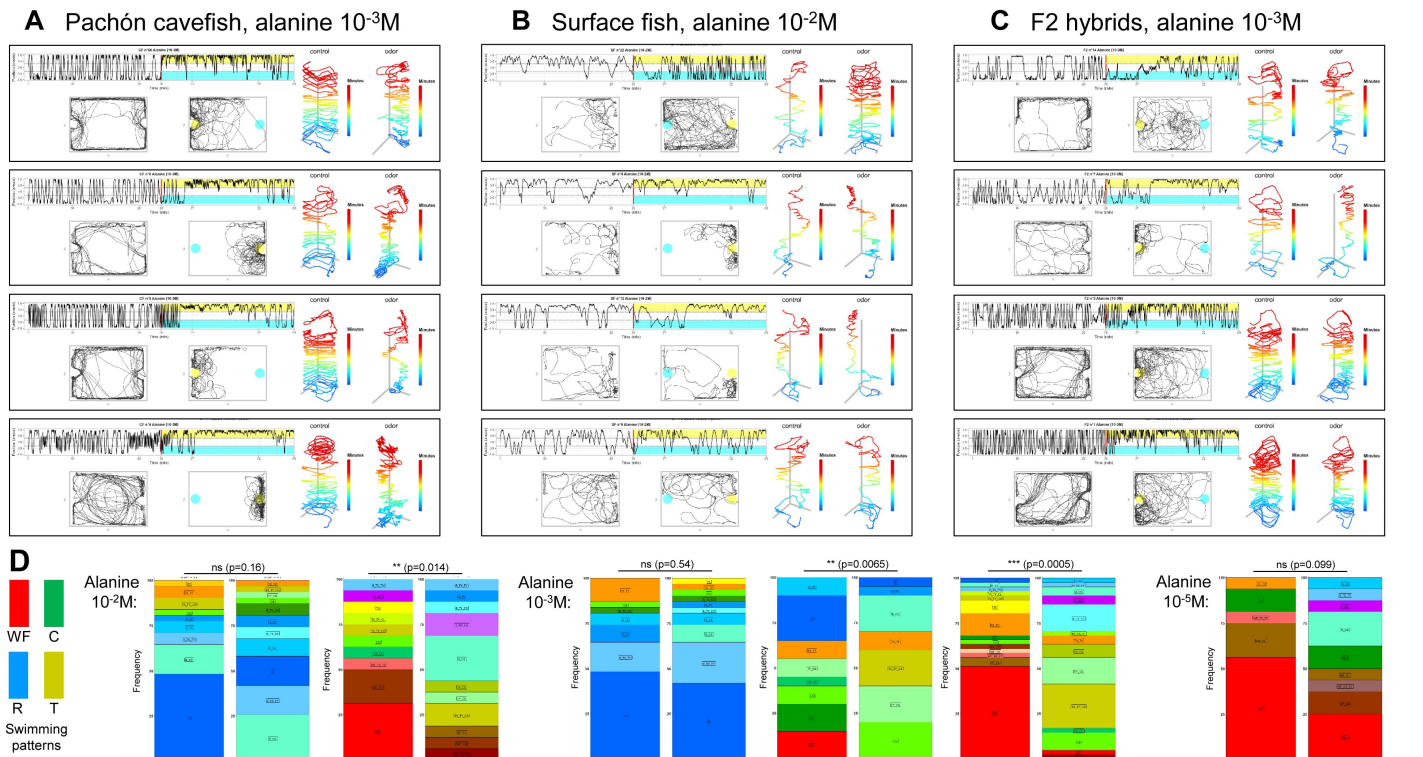
AA', Distribution of swimming patterns exhibited by SF, F2 and CF (number of fish tested is indicated) along 4 testing days, with 24 hours habituation time (**A**). Fisher's exact tests. The color code for the different swimming patterns and combinations is given in the inset below. The table in **A'** shows statistics summary for paired comparison of individual's mean position, swimming speed and the total number of round trips in X plus Y over 4 testing days, using Friedman tests. The results of additional pairwise Wilcoxon comparisons between different days are as follows. All these pairwise comparisons were NS except for CF roundtrip numbers (d1/d2 $p=0.04$; d1/d3 $p=0.0009$; all other comparisons NS) and F2 position (d1/d3 $p=0.022$; d3/d4 $p=0.042$). These results suggest that fish display individual swimming behaviors that are stable through time and may be considered as individual swimming personality or temperament.

B, Lack of change in swimming pattern displayed by SF, CF and F2, before and after water injection on both sides. Fisher's exact tests.

C, Global lack of change in quantitative swimming kinetics in SF (blue), F2 (green) and CF (red), before and after water injection on both side. Mean speed and mean position are shown.

D, Lack of change in quantitative swimming kinetics in SF, CF and F2 when no injection is performed at all. Mean speed and mean position are shown. The result suggests that an individual's behavior is stable throughout the one-hour test.

E, Lack of change in swimming speed or position on the box when SF are recorded in the dark (dark blue), in the light (light blue), or when the light is switched off after 30 minutes.



Suppl Fig2 (with Figure 4).

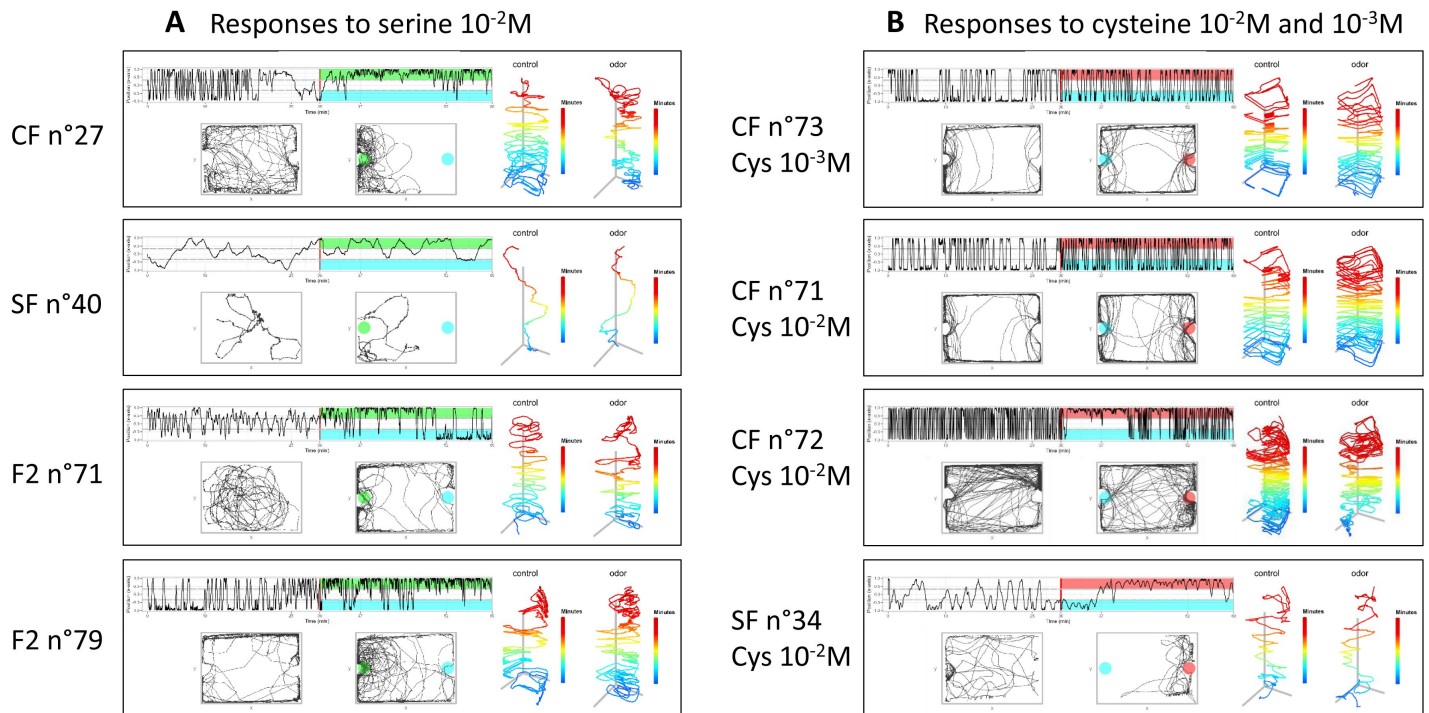
Additional examples of behavioral responses to alanine

A, Examples of CF responses to alanine 10^{-3}M (intermediate concentration). Note the reproducible switch in position toward the odorant side in the box (yellow shading) and the change in behavioral pattern. Note that CF n°8, which was tested twice with the same concentration of alanine, switched from wall following to small circles and thigmotaxis along the Y border of the box in both cases. CF n°4 on the other hand swam in large circles before injection, and changed its pattern to display intense thigmotaxis accompanied with a few circles.

B, Examples of SF responses to alanine 10^{-2}M (high concentration). Upon injection, SF show changes in swimming patterns and increase their swimming speed, suggesting that the stimulus was perceived. SF n°22 swam very actively randomly. SF n°4 switched from random with small circles to circles plus thigmotaxis. SF n°12 added thigmotaxis to its random pattern upon stimulus. Some SF, like SF n°8, did not seem to respond, even at such high alanine concentration.

C, Examples of F2 hybrids responses to alanine 10^{-3}M (intermediate concentration). F2 n°14 did not appear to move its position to the odorant part of the box, but a striking change in swimming pattern from intense thigmotaxis to random swim was observed. F2 n°7 explored close to the odor source and decreased swimming speed. F2 n°3 and 1 switched from wall following to circles with thigmotaxis associated to a change in position towards the odor side.

D, Change in swimming patterns displayed by SF, F2 and CF after injection of alanine 10^{-2}M , 10^{-3}M and 10^{-5}M . Fisher's exact tests.

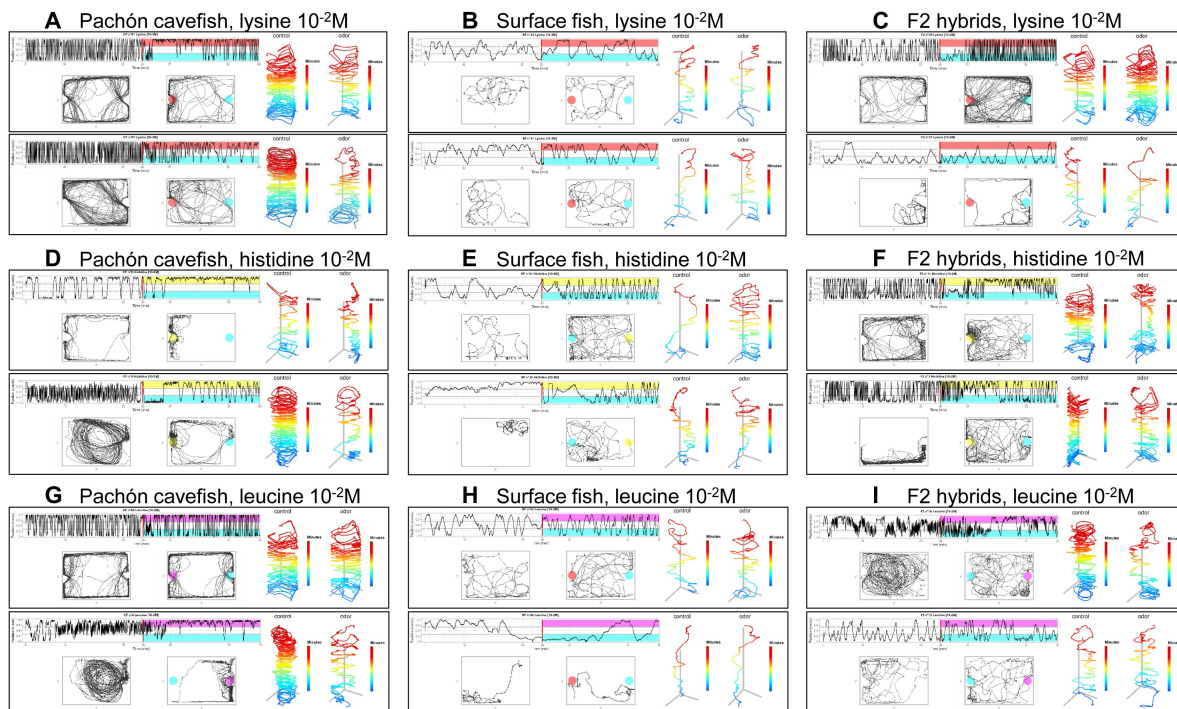


Suppl Fig3 (with Figure 5).

Additional examples of behavioral responses to serine and cysteine

A, responses to serine 10^{-2} M (high concentration). Note the change in behavioral pattern and position in the box observed in CF and F2 but not in SF. The two F2 shown had different baseline swimming patterns: circles in F2 n°71, thigmotaxis in F2 n°79. Yet, both individuals showed a change in swim pattern and an attraction to the serine side of the box.

B, responses to cysteine 10^{-2} M (high concentration) and 10^{-3} M (intermediate concentration). CF n°73 gives an example of an individual that did not respond to 10^{-3} M cysteine: no change in position, no change in pattern, no change in swim kinetics. CF n°71 and CF n°72 on the other hand did respond to cysteine 10^{-2} M, the former by an increase in activity, the later by a transient change in position. SF n°34 is a rare example of a SF that was attracted to cysteine.



Suppl Fig4 (with Figure 6).

Examples of behavioral responses to lysine (ABC), histidine (DEF) and leucine (GHI) 10^{-2} M (high concentration)

In each panel, in the two left graphs, the blue color indicates the water/control injection side. The lysine injection side is shown in red, the histidine injection side is yellow, the leucine injection side is pink.

A, The 2 CF individuals shown decrease their swimming activity upon injection. Moreover, n°81 and 87 shift their position to the odor side of the box. A change in swimming pattern from intensive wall following to thigmotaxis with circles and some random swimming is also observed.

B, The 2 SF individuals shown do not change their swimming pattern (Random) or kinetics in response to lysine.

C, F2 n°35 increased swimming activity upon lysine injection but did not show attraction to odorant side. F2 n° 37 kept its random pattern of swimming and was not attracted to the odorant side.

D, Whether the cavefish individual was initially displaying thigmotaxis (CF n°8) or swimming in large circles (CF n°4), the response to histidine included decreasing the swimming activity and restricting locomotion to the odorant side of the box.

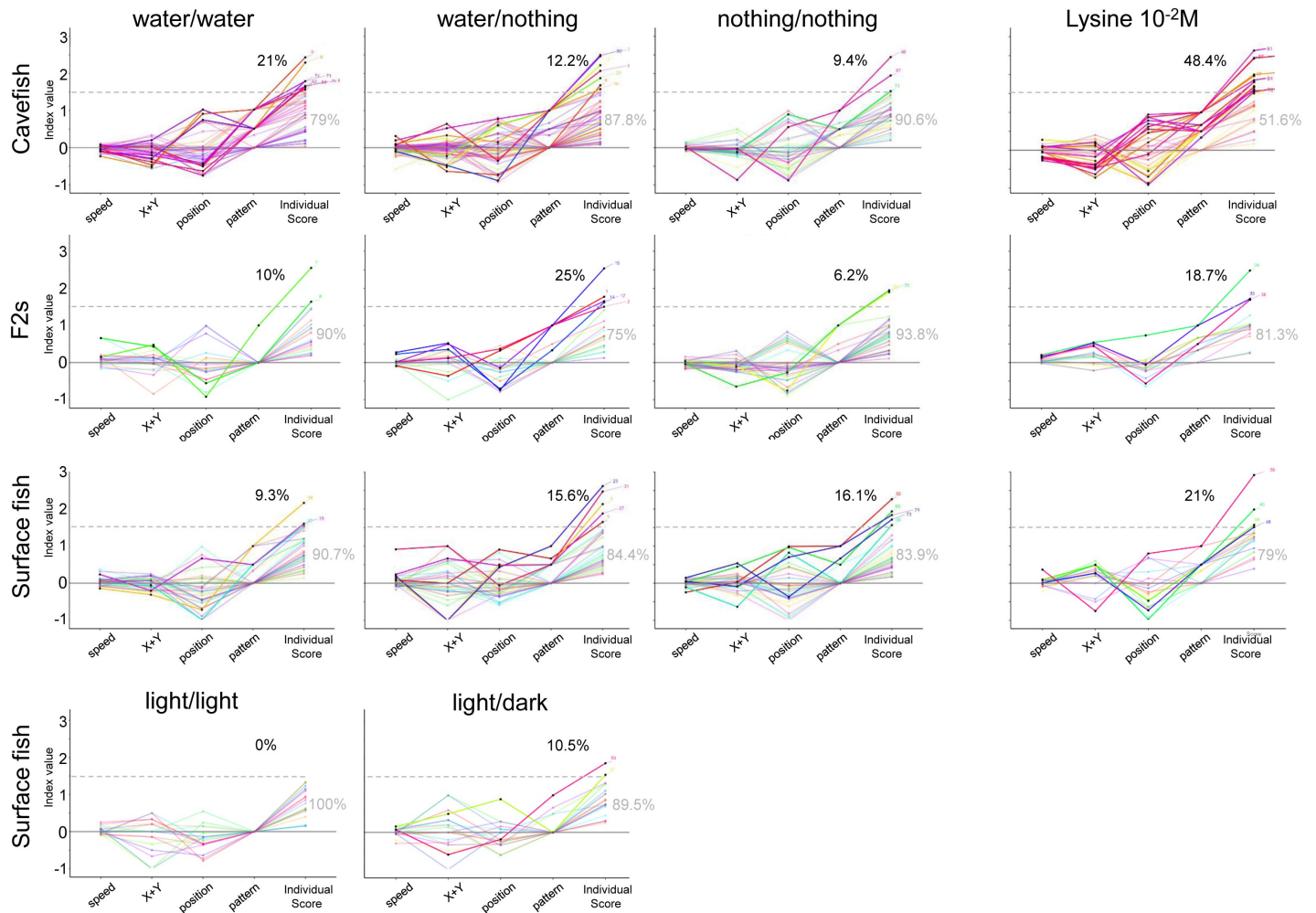
E, The two surface fish individuals shown swam slowly and randomly as baseline pattern. Both increased their activity and spent more time in the odorant side.

F, F2 hybrids n°11 and n°1 were both attracted to histidine odorant side and they changed their swimming pattern.

G, CF n°65 showed no response, neither in activity, in position nor in pattern. CF n°4 drastically changed all swimming parameters.

H, Surface fish showed little reaction to leucine. SF n°43 and n°40 persisted in random swimming patterns, without change of position or swimming kinetics.

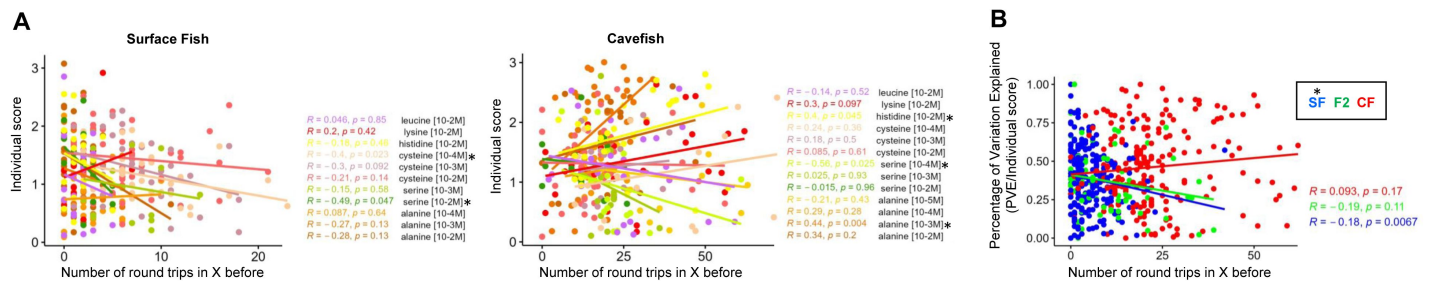
I, F2 hybrids n°16 and n°13 had markedly different baseline patterns (large circles *versus* random swim, respectively). The former changed its pattern to less active and small circles, suggesting that it perceived leucine injection. The latter showed no behavioral response.



Suppl Fig5 (with Figure 7).

Individual olfactory scores of SF, CF and F2 in control and experimental conditions.

The graphs represent index values for the 4 response parameters used to calculate the total individual olfactory score for each fish. Each fish is depicted by a colored line linking its 4 different indexes and its final individual olfactory score. Experimental conditions and fish types are indicated. The threshold value for olfactory score was set at 1.5 (dotted line), because very few individuals of the 3 fish types have scores above this value in control experiments when water is perfused in the box on both sides. The percentages in black and grey indicate the proportion of responders/non-responders, respectively. The colored lines of responders are bright, those of non-responders are pale.



Suppl Fig6 (with Figure 8).

Relationships between the number of round trip in X and the olfactory score

A, Regressions to explore the correlation between baseline number of round trips in X displayed by individual fish before odor injection and their individual olfactory score. Fish and amino acid type and concentrations are indicated. Linear correlations were calculated with Spearman's rank correlation test followed by Student test for the p values. Conditions for which significant correlation was found are labelled with an asterisk.

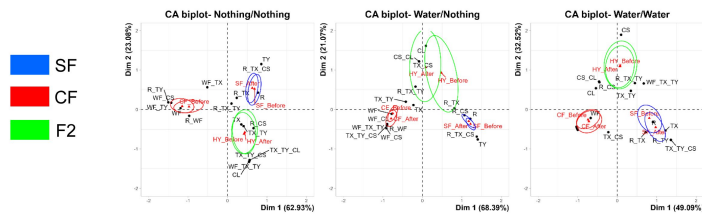
B, The PVE (Percentage of Variation Explained) on the individual score is plotted as a function of individual swimming parameter (numbers of round trips in X), to probe the predictability of the behavioral output as a function of "locomotor activity personality". Linear correlations were calculated with Spearman's rank correlation test followed by Student test for the p values. Conditions for which significant correlation was found are labelled with an asterisk. SF (blue), F2 (green) and CF (red).

WF_TY_CS	
WF_TY	
WF_TX_TY	
WF_TX	
WF_CS_CL	
WF_CS	
WF_CL	
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TX_TY_CL	
TX_TY	
TX_CS_CL	
TX_CS	
TX_CL	
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R_WF_TX	
R_WF_CS	
R_WF_CL	
R_WF	
R_TY_CS	
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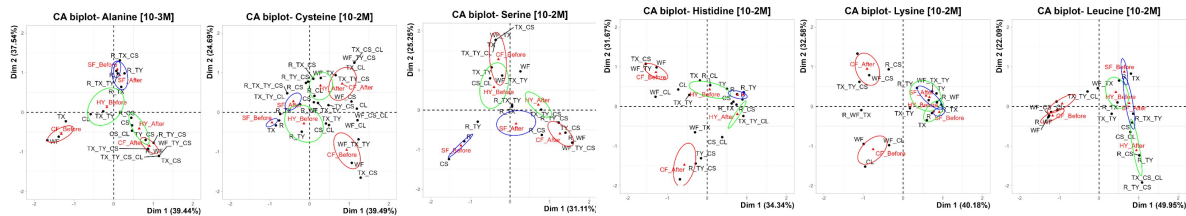
[Suppl Fig7 \(with all figures\).](#)

Complete color code for the description of baseline swimming patterns.

A - controls



B- odors



Suppl Fig8 (with all figures).

Correspondence Analyses (CA) for assessment of behavioral pattern change after different stimuli.

Confident ellipses for each condition in the behavioral space are shown in blue for SF, green for F2 hybrids, red for CF. The conditions are indicated at the top of the graphs.

A, Control experiments. CA shows that behavioral patterns in the 3 types of fish are clearly distinctive in the behavioral space: confidence ellipses are strictly non-overlapping. Moreover, CA confirms that for the 3 types of fish, the swimming pattern does not vary along the one hour test when no injection is made (nothing/nothing), when water is injected on one side (water/nothing), or when water is injected on both sides ((water/water). In all cases, confidences ellipses overlap in the “before” and “after” condition. These analyses fully confirm the results shown in [Figures 2B](#) and [3B](#) with the colored bar plot representation.

B, Odor stimulation experiments. Changes in behavioral patterns in the behavioral space are illustrated for the 6 different amino acids tested. Overlap of confidence ellipses indicates a lack of change in behavioral pattern after odor stimulus, whereas separated confidence ellipses indicate a change in behavioral pattern after odor stimulus. The CA fully confirms the results shown in Table 1 and [Figures 4](#), [5](#), [6](#) with the colored bar plot representation.

References

- Agnès F., Torres-Paz J., Michel P., Rétaux S (2022) **A 3D molecular map of the cavefish neural plate illuminates eye-field organization and its borders in vertebrates** *Development* **149**
- Axel R (1995) **The molecular logic of smell** *Sci Am* **273**:154–9
- Blin M., Fumey J., Lejeune C., Policarpo M., Leclercq J., Père S., Torres-Paz J., Pierre C., Imarazene B., Rétaux S (2020) **Diversity of Olfactory Responses and Skills in Astyanax Mexicanus Cavefish Populations Inhabiting different Caves** *Diversity* **12**:1–21
- Blin M., Tine E., Meister L., Elipot Y., Bibliowicz J., Espinasa L., Rétaux S (2018) **Developmental evolution and developmental plasticity of the olfactory epithelium and olfactory skills in Mexican cavefish** *Dev Biol* **441**:242–251
- Braubach O. R., Fine A., Croll R. P (2012) **Distribution and functional organization of glomeruli in the olfactory bulbs of zebrafish (Danio rerio)** *J Comp Neurol* **520**:2317–39
- Buck L. B (2000) **The molecular architecture of odor and pheromone sensing in mammals** *Cell* **100**:611–8
- Burguera D. *et al.* (2023) **Expanded olfactory system in ray- finned fishes capable of terrestrial exploration** *BMC Biol* **21**
- Chen B., Mao T., Liu Y., Dai W., Li X., Rajput A. P., Pie M. R., Yang J., Gross J. B., Meegaskumbura M (2022) **Sensory evolution in a cavefish radiation: patterns of neuromast distribution and associated behaviour in Sinocyclocheilus (Cypriniformes: Cyprinidae)** *Proc Biol Sci* **289**
- Choi J. H., Duboue E. R., Macurak M., Chanchu J. M., Halpern M. E (2021) **Specialized neurons in the right habenula mediate response to aversive olfactory cues** *Elife* **10**
- de Busserolles F., Fogg L., Cortesi F., Marshall J. (2020) **The exceptional diversity of visual adaptations in deep-sea teleost fishes** *Semin Cell Dev Biol* **106**:20–30
- Duboué E. R., Keene A, Keen AC, Yoshizawa M, McGaugh SE (2016) **Investigating the evolution of sleep in the Mexican cavefish** *Biology and Evolution of the Mexican Cavefish. Chapter 15* :335–359
- Elipot Y., Hinaux H., Callebert J., Launay J., Blin M., Rétaux S (2014) **A Mutation in the Enzyme MonoAmine Oxidase Explains Part of the Astyanax Cavefish Behavioral Syndrome** *Nature Communications Apr* **10**
- Espinasa L., Bonaroti N., Wong J., Pottin K., Queinnec E., Rétaux S (2017) **Contrasting feeding habits of post-larval and adult Astyanax cavefish** *Subterranean Biology* **21**:1–17
- Friedrich R. W., Yaksi E., Judkewitz B., Wiechert M. T (2009) **Processing of odor representations by neuronal circuits in the olfactory bulb** *Ann N Y Acad Sci* **1170**:293–7
- Greenacre M (2021) **Compositional Data Analysis** *Annual Review of Statistics and Its Application* **8**:271–299

- Hara T. J (1994) **Olfaction and gustation in fish: an overview** *Acta Physiol Scand* **152**:207–17
- Helfman G. S., Collette B. B., Facey D. E., Bowen B. W. (2009) **The diversity of fishes. Biology, Evolution, and Ecology**
- Hinaux H., Devos L., Bibliowicz J., Elipot Y., Alié A., Blin M., Rétaux S (2016) **Sensory evolution in blind cavefish is driven by early events during gastrulation and neurulation** *Development* **143**:4521–4532
- Hinaux H., Rétaux S., Elipot Y., Keen AC, Yoshizawa M, McGaugh SE (2016) **Social behavior and aggressiveness in *Astyanax*** *Biology and Evolution of the Mexican Cavefish. Chapter 17* :335–359
- Hubbard P. C., Barata E. N., Ozorio R. O., Valente L. M., Canario A. V (2011) **Olfactory sensitivity to amino acids in the blackspot sea bream (*Pagellus bogaraveo*): a comparison between olfactory receptor recording techniques in seawater** *J Comp Physiol A Neuroethol Sens Neural Behav Physiol* **197**:839–49
- Hüppop K (1987) **Food-finding ability in cave fish (*Astyanax fasciatus*)** *Int. J. Speleol* **16**:59–66
- Iwashita M., Yoshizawa M (2021) **Social-like responses are inducible in asocial Mexican cavefish despite the exhibition of strong repetitive behavior** *Elife* **10**
- Kalueff A. V. *et al.* (2013) **Towards a comprehensive catalog of zebrafish behavior 1.0 and beyond** *Zebrafish* **10**:70–86
- Kermen F., Franco L. M., Wyatt C., Yaksi E (2013) **Neural circuits mediating olfactory-driven behavior in fish** *Front Neural Circuits* **7**
- Kiyohara S., Sakata Y., Yoshitomi T., Tsukahara J (2002) **The 'goatee' of goatfish: innervation of taste buds in the barbels and their representation in the brain** *Proc Biol Sci* **269**:1773–80
- Koide T., Miyasaka N., Morimoto K., Asakawa K., Urasaki A., Kawakami K., Yoshihara Y (2009) **Olfactory neural circuitry for attraction to amino acids revealed by transposon-mediated gene trap approach in zebrafish** *Proc Natl Acad Sci U S A* **106**:9884–9
- Kowalko J (2020) **Utilizing the blind cavefish *Astyanax mexicanus* to understand the genetic basis of behavioral evolution** *J Exp Biol* **223**
- Krubitzer L., Campi K. L., Cooke D. F (2011) **All rodents are not the same: a modern synthesis of cortical organization** *Brain Behav Evol* **78**:51–93
- Li J., Mack J. A., Souren M., Yaksi E., Higashijima S., Mione M., Fetcho J. R., Friedrich R. W (2005) **Early development of functional spatial maps in the zebrafish olfactory bulb** *J Neurosci* **25**:5784–95
- Lunsford E. T., Paz A., Keene A. C., Liao J. C (2022) **Evolutionary convergence of a neural mechanism in the cavefish lateral line system** *Elife* **11**
- Marandel L., Plagnes-Juan E., Marchand M., Callet T., Dias K., Terrier F., Pere S., Vernier L., Panserat S., Rétaux S (2020) **Nutritional regulation of glucose metabolism-related genes in the emerging teleost model Mexican tetra surface fish: a first exploration** *R Soc Open Sci* **7**
- Mathuru A. S., Kibat C., Cheong W. F., Shui G., Wenk M. R., Friedrich R. W., Jesuthasan S (2012) **Chondroitin fragments are odorants that trigger fear behavior in fish** *Curr Biol* **22**:538–44

- Patton P., Windsor S., Coombs S (2010) **Active wall following by Mexican blind cavefish (*Astyanax mexicanus*)** *J Comp Physiol A Neuroethol Sens Neural Behav Physiol* **196**:853–67
- Pierre C., Pradère N., Froc C., Ornelas-Garcia P., Callebert J., Rétaux S (2020) **A mutation in monoamine oxidase (MAO) affects the evolution of stress behavior in the blind cavefish *Astyanax mexicanus*** *J Exp Biol* **223**
- Policarpo M., Bemis K. E., Laurenti P., Legendre L., Sandoz J. C., Rétaux S., Casane D (2022) **Coevolution of the olfactory organ and its receptor repertoire in ray- finned fishes** *BMC Biol* **20**
- Pottin K., Hinaux H., Rétaux S (2011) **Restoring eye size in *Astyanax mexicanus* blind cavefish embryos through modulation of the Shh and Fgf8 forebrain organising centres** *Development* **138**:2467–76
- R Development Core Team (2016) **R: a language and environment for statistical computing**
- Réale D., Reader S. M., Sol D., McDougall P. T., Dingemanse N. J (2007) **Integrating animal temperament within ecology and evolution** *Biol Rev Camb Philos Soc* **82**:291–318
- Rey S., Jin X., Damsgard B., Begout M. L., Mackenzie S (2021) **Analysis across diverse fish species highlights no conserved transcriptome signature for proactive behaviour** *BMC Genomics* **22**
- Schemmel C (1967) **Vergleichende Untersuchungen an den Hautsinnesorganen ober- and unter-irdisch lebender *Astyanax*-Foramen** *Z Morph Tiere* **61**:255–316
- Sharma S., Coombs S., Patton P., de Perera Burt (2009) **The function of wall-following behaviors in the Mexican blind cavefish and a sighted relative, the Mexican tetra (*Astyanax*)** *J Comp Physiol A Neuroethol Sens Neural Behav Physiol* **195**:225–40
- Simon V., Hyacinthe C., Rétaux S (2019) **Breeding behavior in the blind Mexican cavefish and its river-dwelling conspecific** *PLoS One* **14**
- Sylvester J. B., Rich C. A., Loh Y. H., van Staaden M. J., Fraser G. J., Streelman J. T. (2010) **Brain diversity evolves via differences in patterning** *Proc Natl Acad Sci U S A* **107**:9718–23
- Torres-Paz J., Leclercq J., Rétaux S (2019) **Maternally regulated gastrulation as a source of variation contributing to cavefish forebrain evolution** *Elife* **8**
- Tricas T. C., Kajiura S. M., Summers A. P (2009) **Response of the hammerhead shark olfactory epithelium to amino acid stimuli** *J Comp Physiol A Neuroethol Sens Neural Behav Physiol* **195**:947–54
- Varatharasan N., Croll R. P., Franz-Odenaal T (2009) **Taste bud development and patterning in sighted and blind morphs of *Astyanax mexicanus*** *Dev Dyn* **238**:3056–64
- Vitebsky A., Reyes R., Sanderson M. J., Michel W. C., Whitlock K. E (2005) **Isolation and characterization of the laque olfactory behavioral mutant in the zebrafish, *Danio rerio*** *Dev Dyn* **234**:229–42
- Wakisaka N., Miyasaka N., Koide T., Masuda M., Hiraki-Kajiyama T., Yoshihara Y (2017) **An Adenosine Receptor for Olfaction in Fish** *Curr Biol* **27**:1437–1447

Yamamoto Y., Byerly M. S., Jackman W. R., Jeffery W. R (2009) **Pleiotropic functions of embryonic sonic hedgehog expression link jaw and taste bud amplification with eye loss during cavefish evolution** *Dev Biol* **330**:200–11

Yoshihara Y (2008) **Molecular Genetic Dissection of the Zebrafish Olfactory System** *Results Probl Cell Differ*

Yoshizawa M, Keen AC, Yoshizawa M, McGaugh SE (2016) **The evolution of sensory adaptation in *Astyanax mexicanus*** *Biology and Evolution of the Mexican Cavefish. Chapter 13* :335–359

Yoshizawa M., Goricki S., Soares D., Jeffery W. R (2010) **Evolution of a behavioral shift mediated by superficial neuromasts helps cavefish find food in darkness** *Curr Biol* **20**:1631–6

Yoshizawa M., Jeffery W. R (2011) **Evolutionary tuning of an adaptive behavior requires enhancement of the neuromast sensory system** *Communicative and Integrative Biology* **4**:89–91

Yoshizawa M., Jeffery W. R., van Netten S. M., McHenry M. J. (2014) **The sensitivity of lateral line receptors and their role in the behavior of Mexican blind cavefish (*Astyanax mexicanus*)** *J Exp Biol* **217**:886–95

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Summary:

The paper explores chemosensory behaviour in surface and cave morphs and F2 hybrids in the Mexican cave fish *Astyanax mexicanus*. The authors develop a new behavioural assay for the long-term imaging of individual fish in a parallel high-throughput setup. The authors first demonstrate that the different morphs show different basal exploratory swimming patterns and that these patterns are stable for individual fish. Next, the authors test the attraction of fish to various concentrations of alanine and other amino acids. They find that the cave morph is a lot more sensitive to chemicals and shows directional chemotaxis along a diffusion gradient of amino acids. Surface fish, although can detect the chemicals, do not show marked chemotaxis behaviour and have an overall lower sensitivity. These differences have been reported previously but the authors report longer-term observations on many individual fish of both morphs and their F2 hybrids. The data also indicate that the observed behaviour is a quantitative genetic trait. The approach presented will allow the mapping of genes contribution to these traits. The work will be of general interest to behavioural neuroscientists and those interested in olfactory behaviours and the individual variability in behavioural patterns.

Strengths:

The authors provide a large dataset of swimming behaviour for surface fish and cave fish and also their F2 hybrids, demonstrating large differences in chemosensory behaviour and indicating that this is a quantitative genetic trait.

One strength of the paper is the development of a new and improved setup for the behavioural imaging of individual fish for extended periods and under chemosensory stimulation. The authors show that cave fish need up to 24 h of habituation to display a behavioural pattern that is consistent and unlikely to be due to the stressed state of the animals. The setup also uses relatively large tanks that allows the build-up of chemical gradients.

With their new system, the authors could generate cleaner results without mechanical disturbances. The authors characterize multiple measurements to score the odour response behaviours and also developed a new personality analysis. Their conclusion that cave fish evolved as a specialist to sense alanine and histidine among 6 tested amino acids was well supported by their data.

Weaknesses:

Further work will be needed to pinpoint the nature of the genetic changes and neurobiological mechanisms that underlie the differences between the two forms and the large individual variation of behaviours.

The authors did not measure the concentrations of alanine and other amino acids in the local cave water and surface water.

<https://doi.org/10.7554/eLife.92861.2.sa0>

Author response:

The following is the authors' response to the original reviews.

eLife assessment

In this important paper, Blin and colleagues develop a high-throughput behavioral assay to test spontaneous swimming and olfactory preference in individual Mexican cavefish larvae. The authors present compelling evidence that the surface and cave morphs of the fish show different olfactory preferences and odor sensitivities and that individual fish show substantial variability in their spontaneous activity that is relevant for olfactory behaviour. The paper will be of interest to neurobiologists working on the evolution of behaviour, olfaction, and the individuality of behaviour.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The authors posed a research question about how an animal integrates sensory information to optimize its behavioral outputs and how this process evolved. Their data (behavioral output analysis with detailed categories in response to the different odors in different concentrations by comparing surface and cave populations and their hybrid) partially answer this tough question. They built a new low-disturbance system to answer the question. They also found that the personality of individual fish is a good predictor of behavioral outputs against odor response. They concluded that cavefish evolved to specialize their response to alanine and histidine while surface fish are more general responders, which was supported by their data.

Strengths:

With their new system, the authors could generate clearer results without mechanical disturbances. The authors characterize multiple measurements to score the odor response behaviors, and also brought a new personality analysis. Their conclusion that cavefish evolved as a specialist to sense alanine and histidine among 6 tested amino acids was well supported by their data.

Weaknesses:

The authors posed a big research question: How do animals evolve the processes of sensory integration to optimize their behavioral outputs? I personally feel that, to answer the questions about how sensory integration generates proper (evolved) behavior, the authors at least need to show the ecological relevance of their response. For the alanine/histidine preference in cavefish, they need data for the alanine and other amino acid concentrations in the local cave water and compare them with those of surface water.

We agree with the reviewer. This is why, in the Discussion section, we had written: "...Such significant variations in odor preferences or value may be adaptive and relate to the differences in the environmental and ecological conditions in which these different animals

live. However, the reason why Pachón cavefish have become “alanine specialists” remains a mystery and prompts analysis of the chemical ecology of their natural habitat. Of note, we have not found an odor that would be repulsive for *Astyanax* so far, and this may relate to their opportunist, omnivorous and detritivore regime (Espinasa et al., 2017; Marandel et al., 2020).” This is also why we currently develop field work projects aimed at clarifying this question. However, such experiments and analyses are challenging, practically and technically. We hope we can reach some conclusions in the future.

To complete the discussion we have also added an important hypothesis: “Alternatively, specialization for alanine may not need to be specific for an olfactory cue present only, or frequently, or in high amounts in caves. Bat guano for example, which is probably the main source of food in the Pachón cave, must contain many amino acids. Enhanced recognition of one of them - in the present case alanine but evolution may have randomly acted for enhanced recognition of another amino acid – should suffice to confer cavefish with augmented sensitivity to their main source of nutriment.”

Also, as for "personality matters", I read that personality explains a large variation in surface fish. Also, thigmotaxis or wall-following cavefish individuals are exceeded to respond well to odorants compared with circling and random swimming cavefish individuals. However, I failed to understand the authors' point about how much percentages of the odorant-response variations are explained (PVE) by personality. Association (= correlation) was good to show as the authors presented, but showing proper PVE or the effect size of personality to predict the behavioral outputs is important to conclude "personality is matter"; otherwise, the conclusion is not so supported.

From the above, I recommend the authors reconsider the title also their research questions well. At this moment, I feel that the authors' conclusions and their research questions are a little too exaggerated, with less supportive evidence.

Thank you for this interesting suggestion, which we have fully taken into consideration. We have therefore now calculated and plotted PVE (the percentage of variation explained on the olfactory score) as a function of swimming speed or as a function of swimming pattern. The results are shown in modified Figure 8 of our revised ms and they suggest that the personality (here, swimming patterns or swimming speed) indeed predicts the olfactory response skills. Therefore, we would like to keep our title as we provide support for the fact that “personality matters”.

Also, for the statistical method, Fisher's exact test is not appropriate for the compositional data (such as Figure 2B). The authors may quickly check it at https://en.wikipedia.org/wiki/Compositional_data or <https://www.annualreviews.org/doi/pdf/10.1146/annurev-statistics-042720-124436>.

The authors may want to use centered log transformation or other appropriate transformations (Rpackage could be: <https://doi.org/10.1016/j.cageo.2006.11.017>). According to changing the statistical tests, the authors' conclusion may not be supported.

Actually, in most cases, the distributions are so different (as seen by the completely different colors in the distribution graphs) that there is little doubt that swimming behaviors are indeed different between surface and cavefish, or between ‘before’ and ‘after’ odor stimulation. However, it is true that Fisher’s exact test is not fully appropriate because data can be considered as compositional type. For this kind of data, centered log transformation have been suggested. However, our dataset contains many zeros, and this is a case where log transformations have difficulty handling.

To help us dealing with our data, the reviewer proposed to consider the paper by Greenacre (2021) (<https://www.annualreviews.org/doi/pdf/10.1146/annurev-statistics-042720-124436>). In his paper, Greenacre clearly wrote: "Zeros in compositional data are the Achilles heel of the logratio approach (LRA)."

Therefore, we have now tested our data using CA (Correspondence Analysis), that can deal with table containing many zeros and is a trustable alternative to LRA (Cook-Thibeuau, 2021; Greenacre, 2011).

The results of CA analysis are shown in Supplemental figure 8 and they fully confirm the difference in baseline swimming patterns between morphs as well as changes (or absence of changes) in behavioral patterns after odor stimulation suggested by the colored bar plots in main figures, with confidence ellipses overlapping or not overlapping, depending on cases. Therefore, the CA method fully confirms and even strengthens our initial interpretations.

Finally, we have kept our initial graphical representation in the ms (color-coded bar plots; the complete color code is now given in Suppl. Fig7), and CA results are shown in Suppl. Figure 8 and added in text.

Reviewer #2 (Public Review):

*In their submitted manuscript, Blin et al. describe differences in the olfactory-driven behaviors of river-dwelling surface forms and cave-dwelling blind forms of the Mexican tetra, *Astyanax mexicanus*. They provide a dataset of unprecedented detail, that compares not only the behaviors of the two morphs but also that of a significant number of F2 hybrids, therefore also demonstrating that many of the differences observed between the two populations have a clear (and probably relatively simple) genetic underpinning.*

*To complete the monumental task of behaviorally testing 425 six-week-old *Astyanax* larvae, the authors created a setup that allows for the simultaneous behavioral monitoring of multiple larvae and the infusion of different odorants without introducing physical perturbations into the system, thus biasing the responses of cavefish that are particularly fine-tuned for this sensory modality. During the optimization of their protocol, the authors also found that for cave-dwelling forms one hour of habituation was insufficient and a full 24 hours were necessary to allow them to revert to their natural behavior. It is also noteworthy that this extremely large dataset can help us see that population averages of different morphs can mask quite significant variations in individual behaviors.*

Testing with different amino-acids (applied as relevant food-related odorant cues) shows that cavefish are alanine- and histidine-specialists, while surface fish elicit the strongest behavioral responses to cysteine. It is interesting that the two forms also react differently after odor detection: while cave-dwelling fish decrease their locomotory activity, surface fish increase it. These differences are probably related to different foraging strategies used by the two populations, although, as the observations were made in the dark, it would be also interesting to see if surface fish elicit the same changes in light as well.

Thank you for these nice comments.

Further work will be needed to pinpoint the exact nature of the genetic changes that underlie the differences between the two forms. Such experimental work will also reveal how natural selection acted on existing behavioral variations already present in the SF population.

Yes. Searching for genetic underpinnings of the sensory-driven behavioral differences is our current endeavor through a QTL study and we should be able to report it in the near future.

It will be equally interesting, however, to understand what lies behind the large individual variation of behaviors observed both in the case surface and cave populations. Are these differences purely genetic, or perhaps environmental cues also contribute to their development? Does stochasticity provided by the developmental process has also a role in this? Answering these questions will reveal if the evolvability of Astyanax behavior was an important factor in the repeated successful colonization of underground caves.

Yes. We will also access (at least partially) responses to most of these questions in our current QTL study.

Reviewer #3 (Public Review):

Summary:

*The paper explores chemosensory behaviour in surface and cave morphs and F2 hybrids in the Mexican cavefish *Astyanax mexicanus*. The authors develop a new behavioural assay for the longterm imaging of individual fish in a parallel high-throughput setup. The authors first demonstrate that the different morphs show different basal exploratory swimming patterns and that these patterns are stable for individual fish. Next, the authors test the attraction of fish to various concentrations of alanine and other amino acids. They find that the cave morph is a lot more sensitive to chemicals and shows directional chemotaxis along a diffusion gradient of amino acids. For surface fish, although they can detect the chemicals, they do not show marked chemotaxis behaviour and have an overall lower sensitivity. These differences have been reported previously but the authors report longer-term observations on many individual fish of both morphs and their F2 hybrids. The data also indicate that the observed behavior is a quantitative genetic trait. The approach presented will allow the mapping of genes' contribution to these traits. The work will be of general interest to behavioural neuroscientists and those interested in olfactory behaviours and the individual variability in behavioural patterns.*

Strengths:

A particular strength of this paper is the development of a new and improved setup for the behavioural imaging of individual fish for extended periods and under chemosensory stimulation. The authors show that cavefish need up to 24 h of habituation to display a behavioural pattern that is consistent and unlikely to be due to the stressed state of the animals. The setup also uses relatively large tanks that allow the build-up of chemical gradients that are apparently present for at least 30 min.

The paper is well written, and the presentation of the data and the analyses are clear and to a high standard.

Thank you for these nice comments.

Weaknesses:

One point that would benefit from some clarification or additional experiments is the diffusion of chemicals within the behavioural chamber. The behavioural data suggest that the chemical gradient is stable for up to 30 min, which is quite surprising. It would be great if the authors could quantify e.g. by the use of a dye the diffusion and stability of chemical gradients.

OK. We had tested the diffusion of dyes in our previous setup and we also did in the present one (not shown). We think that, due to differences of molecular weight and hydrophobicity between the tested dyes and the amino acid molecules we are using, their diffusion does not constitute a proper read-out of actual amino acid diffusion. We anticipate that amino acid diffusion is extremely complex in the test box, possibly with odor plumes diffusing and evolving in non-gradient patterns, in the 3 dimensions of the box, and potentially further modified by the fish swimming through it, the flow coming from the opposite water injection side and the borders of the box. This is the reason why we have designed the assay with contrasting “odor side” and “water control side”. Moreover, our question here is not to determine the exact concentration of amino acid to which the fish respond, but to compare the responses in cavefish, surface fish and F2 hybrids. Finally and importantly, we have performed dose/response experiments whereby varying concentrations have been presented for 3 of the 6 amino acids tested, and these experiments clearly show a difference in the threshold of response of the different morphs.

The paper starts with a statement that reflects a simplified input-output (sensory-motor) view of the organisation of nervous systems. "Their brains perceive the external world via their sensory systems, compute information and generate appropriate behavioral outputs." The authors' data also clearly show that this is a biased perspective. There is a lot of spontaneous organised activity even in fish that are not exposed to sensory stimulation. This sentence should be reworded, e.g. "The nervous system generates autonomous activity that is modified by sensory systems to adapt the behavioural pattern to the external world." or something along these lines.

Done

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

In addition to my comments in the "weakness" section above, here are my other comments.

How many times fish were repeatedly assayed and what the order (alanine followed by cysteine, etc) was, is not clear (Pg 24, Materials and Methods). I am afraid that fish memorize the prior experience to get better/worse their response to the higher conc of alanine, etc. Please clarify this point.

Many fish were tested in different conditions on consecutive days, indeed. Most often, control experiments (eg, water/nothing; water/water; nothing/nothing) were followed by odor testing. In such cases, there is no risk that fish memorize prior experience and that such previous experience interferes with response to odor. In other instances, fish were tested with a low concentration of one amino acid, followed by a high concentration of another amino acid, which is also on the safe side. Of note, on consecutive days, the odors were always perfused on alternate sides of the test box, to avoid possibility of spatial memory. Finally, in the few cases where increasing concentrations of the same amino acids were perfused consecutively, 1) they were perfused on alternate sides, 2) if the fish does not detect a low concentration below threshold / does not respond, then prior experience should not interfere for responding to higher concentrations, and 3) we have evidence (unpublished, current studies) that when a fish is given increasing concentrations of the same amino acid above detection threshold, then the behavioral response is stable and reproducible (eg does not decrease or increase).

Minor points:

Thygmotaxis and wall following.

Classically, thigmotaxis and wall following are treated as the same (sharma et al., 2009; <https://pubmed.ncbi.nlm.nih.gov/19093125/>) but the authors discriminate it in thigmotaxis at X-axis and Y-axis because fish repeatedly swam back and forth on x-axis wall or y-axis wall. I understand the authors' point to discriminate WF and T but present them with more explanations (what the differences between them) in the introduction and result sections.

Done

Pg5 "genetic architecture" in the introduction.

"Genetic architecture" analysis needs a more genomic survey, such as GWAS, QTL mapping, and Hi-C. Phenotype differences in F2 generation can be stated as "genetic factor(s)" "genetic component(s)", etc. please revise.

Done

Pg10 At the serine treatment, the authors concluded that "...suggesting that their detection threshold for serine is lower than for alanine." I believe that the 'threshold for serine is higher' according to the authors' data. Their threshold-related statement is correct in Pg21 "as SF olfactory concentration detection threshold are higher than CF,..." So the statement on page 10 is a just mistake, I think. Please revise.

Done (mistake indeed)

Pg11 After explaining Fig5, the statement "In sum, the responses of the different fish types to different concentrations of different amino acids were diverse and may reflect complex, case-by-case, behavioral outputs" does not convey any information. Please revise.

OK. Done : "In sum, the different fish types show diverse responses to different concentrations of different amino acids."

For the personality analysis (Fig 7)

The index value needs more explanation. I read the materials and methods three times but am still confused. From the equation, the index does not seem to exceed 1.0, unless the "before score" was a negative value, and the "after score" value was positive. I could not get why the authors set a score of 1.5 as the threshold for the cumulative score of these different behavior index values (= individual score). Please provide more description. Currently, I am skeptical about this index value in Fig 7.

Done, in results and methods.

Pg15 the discussion section

Please discuss well the difference between the authors' finding (cavefish respond $10^{-4}M$ for position and surface fish responded 10^{-4} for thig-Y; Fig 4AB), and those in Hinaux et al. 2016 (cavefish responded $10^{-10}M$ alanine but surface fish responded $10^{-5}M$ or higher). It seems that surface fish could respond to the low conc of alanine as cavefish do, which is opposed to the finding in Hinaux 2016.

The increase in NbrtY at population level for surface fish with 10-4M alanine (~10-6M in box) was most probably due to only a few individuals. Contrarily to cavefish, all other parameters were unchanged in surface fish for this concentration. Moreover, at individual level, only 3.2% of surface fish had significant olfactory scores (to be compared to 81.3% for cavefish). Thus, we think that globally this result does not contradict our previous findings in Hinaux et al (2016), and solely represent the natural, unexplained variations inherent to the analysis of complex animal behaviors – even when we attempt to use the highest standards of controlled conditions.

Of note, in the revised version, we have now included a full dose/response analysis for alanine concentration ranging from 10-2M to 10-10M, on cavefish. Alanine 10-5M has significant effects (now shown in Suppl Fig2 and indicated in text; a column has been added for 10-5M in Summary Table 1). Lower concentrations have milder effects (described in text) but confirm the very low detection threshold of cavefish for this amino acid.

Pg19, "In sum, CF foraging strategy has evolved in response to the serious challenge of finding food in the dark"

My point is the same as explained in the 'weakness' section above: how this behavior is effective in the cave life, if they conclude so? Please explain or revise this statement.

The present manuscript reports on experiments performed in “artificial” and controlled laboratory conditions. We are fully aware that these conditions are probably distantly related to conditions encountered in the wild. Note that we had written in original version (page 20) “...for 6-week old juveniles in a rectangular box - but the link may be more elusive when considering a fish swimming in a natural, complex environment.” As the reviewer may know, we also perform field studies in a more ethological approach of animal behaviors, thus we may be able to discuss this point more accurately in the future.

Pg20 "To our knowledge, this is the first time individual variations are taken into consideration in Astyanax behavioral studies."

This is wrong. Please see Fernandes et al., 2022. (<https://pubmed.ncbi.nlm.nih.gov/36575431/>).

OK. The sentence is wrong if taken in its absolute sense, i.e., considering inter-individual variations of a given parameter (e.g., number of neuromasts per individual or number of approaches to vibrating rod in Fernandez et al, 2022). In this same sense, Astyanax QTL studies on behaviors in the past also took into account variations among F2 individuals. Here, we wanted to stress that personality was taken into consideration. The sentence has been changed: “To our knowledge, this is the first time individual temperament is taken into consideration in Astyanax behavioral studies.”

Figure 2B and others.

The order of categories (R, R-TX, etc) should match in all columns (SF, F2, and CF). Currently, the category orders seem random or the larger ratio categories at the bottom, which is quite difficult to compare between SF, F2, and CF. Also, the writings in Fig 2A (times, Y-axis labels, etc), and the bargraphs' writings are quite difficult to read in Fig 2B, Fig 3B 4H, 5GN, 6EFG. Also, no need to show fish ID in Fig 2C in the current way, but identify the fish data points of the fish in Fig 2D (SF#40, CF#65, and F2#26) in Fig 2C if the authors want to show fish ID numbers in the boxplots. Fish ID numbers in other boxplot figures are recommended to be removed too.

We have thought a lot on how to best represent the distributions of swimming patterns in graphs such as Fig 2B and others. The difficulty is due to the existence of many combinations (33 possibilities in total, see new Suppl Fig7), which are never the same in different plots/conditions because individual tested fish are different. We decided that that the best way was to represent, from bottom to top, the most used to the less used swimming patterns, and to use a color code that matches at best the different combinations. It was impossible to give the full color code on each figure, therefore it was simplified, and we believe that the results are well conveyed on the graphs. We would like to keep it as it is. To respond (partially) to the reviewer's concern, we have now added a full color code description in a new Supplemental Figure 7 (associated to Methods).

Size of lettering has been modified in all pattern graphs like Fig2A. Thanks for the suggestion, it reads better now.

Finally, we would like to keep the fish ID numbers because this contributes to conveying the message of the paper, that individuality matters.

Raw data files were not easy to read in Excel or LibreOffice. Please convert them into the csv format to support the rigor in the authors' conclusion.

We do not understand this request. Our very large dataset must be analysed with R, not excel for stats or for plotting and pattern analysis. However, raw data files can be opened in excel with format conversion.

Reviewer #2 (Recommendations For The Authors):

I think most of the experimental procedures (with few exceptions, see below) are well-defined and nicely described, so the majority of my suggestions will be related to the visualization of the data. I think the authors have done a great job in presenting this complex dataset, but there are still some smaller tweaks that could be used to increase the legibility of the presented data.

First and perhaps foremost, a better definition of the swimming pattern subsets is needed. I have no problem understanding the main behavioral types, but whereas the color codes for these suggest that there is continuous variance within each pattern, it is not clear (at least to me), what particular aspect(s) of the behaviors vary. Also, whereas the sidebars/legends suggest a continuum within these behaviors, the bar charts themselves clearly present binned data. I did not find a detailed description of how the binning was done. As this has been - according the Methods section - a manual process, more clarity about the details of the binning would be welcome. I would also suggest using binned color codes for the legends as well.

Done, in Results and Methods. We hope it is now clear that there is no “continuum”, rather multiple combinations of discrete swimming patterns. The gradient aspect in color code in figures has been removed to avoid the idea of continuum. According to the chosen color code, WF is in red, R in blue, T in yellow and C in green. Then, combination are represented by colors in between, for example, R+WF is purple. We have now added a full color code description for the swimming patterns and their combinations in a new Supplemental Figure 7 (associated to Methods).

Also, to better explain the definition of the swimming patterns and the graphical representation, it now reads (in Methods):

“The determination of baseline swimming patterns and swimming patterns after odor injection was performed manually based on graphical representations such as in Figure 2A

or Figure 3A. Four distinctive baseline behaviors clearly emerged: random swim (R; defined as haphazard swimming with no clear pattern, covering entirely or partly the surface of the arena), wall following (WF; defined as the fish continuously following along the 4 sides of the box and turning around it, in a clockwise or counterclockwise fashion), large or small circles (C; self explanatory), and thigmotaxis (T, along the X- or the Y-axis of the box; defined as the fish swimming back and forth along one of the 4 sides of the box). On graphical representations of swimming pattern distributions, we used the following color code: R in blue, WF in red, C in green, T in yellow. Of note, many fish swam according to combination(s) of these four elementary swimming patterns (see descriptions in the legends of Supplemental figures, showing many examples). To fully represent the diversity and the combinations of swimming patterns used by individual fish, we used an additional color code derived from the “basic” color code described above and where, for example R+WF is purple. The complete combinatorial color code is shown in Suppl. Fig7.”

It would be also easier to comprehend the stacked bar charts, presenting the particular swimming patterns in each population, if the order of different swimming patterns was the same for all the plots (e.g. the frequency of WF always presented at the bottom, R on the top, and C and T in the middle). This would bring consistency and would highlight existing differences between SF, CF, and F2s. Furthermore, such a change would also make it much easier to see (and compare) shifts in behaviors.

We have thought a lot on how to best represent the distributions of swimming patterns in graphs such as Fig 2B and others. The difficulty is due to the existence of many combinations, which are never the same in different plots/conditions because the individual fish tested are different. We decided to keep it as it currently stands, because we think re-doing all the graphs and figures would not significantly improve the representation. In fact, we think that the differences between morphs (dominant blue in SF, dominant red in CF) and between conditions (bar charts next to each other) are easy to interpret at first glance in the vast majority of cases. Moreover, they are now completed by CA analyses (Suppl Figure 8).

While the color coding of the timeline in the "3D" plots presented for individual animals is a nice feature, at the moment it is slightly confusing, as the authors use the same color palette as for the stacked bar charts, representing the proportionality of the particular swimming patterns. As the y-axis is already representing "time" here, the color coding is not even really necessary. If the authors would like to use a color scheme for aesthetic reasons, I would suggest using another palette, such as "grey" or "viridis".

We would like to keep the graphical aspect of our figures as they are, for aesthetic reasons. To avoid confusion with stacked bar chart color code, we have added a sentence in Methods and in the legend of Figure 2, where the colors first appear:

“The complete combinatorial color code is shown in Suppl. Figure 7. Of note, in all figures, the swimming pattern color code does not relate whatsoever with the time color code used in the 2D plus time representation of swimming tracks such as in Figure 2A”.

I would also suggest changing the boxplots to violin-plots. Figure 7 clearly shows bimodality for F2 scores (something, as the authors themselves note, not entirely surprising given the probably polygenic nature of the trait), but looking at SF and CF scores I think there are also clear hints for non-normal distributions. If non-normal distribution of traits is the norm, violin-plots would capture the variance in the data in a more digestible way. (The existence of differently behaving cohorts within the population of both SF and CF forms would also help to highlight the large pre-existing variance, something that was probably exploited by natural selection as well, as mentioned briefly in the Discussion by the authors, too.)

The bimodal distribution of scores shown by F2s in Figure 7B is indeed probably due to the polygenic nature of the trait. However, such distribution is rather the exception than the norm. Moreover, the boxplot representations we have used throughout figures include all the individual points, and outliers can be identified as they have the fish ID number next to them. This allows the reader to grasp the variance of the data. Again, redoing all graphs and figures would constitute a lot of work, for little gain in term of conveying the results. Therefore, we choose not to change the boxplot for violin plots.

The summary data of individual scores in Table 1B shows some intriguing patterns, that warrant a bit further discussion, in my opinion. For example, we can see opposite trends in scores of SF and CF forms with increasing alanine concentration. Is there an easy explanation for this? Also, in the case of serine, the CF scores do not seem to respond in a dose-dependent manner and puzzlingly at 10^{-3} M serine concentration F2 scores are above those of both grandparental populations.

That is true. However, we have no simple explanation for this. To begin responding to this question, we have now performed full dose/responses expts for alanine (concentrations tested from 10-2M to 10-10M on cavefish; confirm that CF are bona fide “alanine specialists”) and for serine (10-2M to 10-4M tested on both morphs; confirm that both morphs respond well to this amino acid). These complementary results are now included in text and figures (partially) and in the summary table 1.

If anything is known about this, I would also welcome some discussion on how thigmotactic behavior, a marker of stress in SF, could have evolved to become the normal behavior of CF forms, with lower cortisol levels and, therefore lower anxiety.

We actually think thigmotactism is a marker of stress in both morphs. See Pierre et al, JEB 2020, Figure S3A: in both SF and CF thigmotaxis behavior decreases after long habituation times. In our hands, the only difference between the two morphs is that surface fish (at 5 month of age) express stress by thigmotactism but also freezing and rapid erratic movements, while cavefish have a more restricted stress repertoire.

This is why in the present paper we have carefully made the distinction between thigmotactism (= possible stress readout) and wall following (= exploratory behavior). Our finding that WF and large circles confers better olfactory response scores to cavefish is in strong support of the different nature of these two swimming patterns. Then, why is swimming along the 4 walls of a tank fundamentally different from swimming along one wall? The question is open, although the number of changes of direction is probably an important parameter: in WF the fish always swims forward in the same direction, while in T the fish constantly changes direction when reaching the corner of the tank – which is similar to erratic swim in stressed surface fish.

Finally two smaller suggestions:

- *When referring to multiple panels on the same figure it would be better to format the reference as "Figure 4D-G" instead of "Figure 4DEFG";*

Done

- *On page 4, where the introduction reads as "although adults have a similar olfactory rosette with 2025 lamellae", in my opinion, it would be better to state that "while adults of the two forms have a similar olfactory rosette with 20-25 lamellae".*

Done

Reviewer #3 (Recommendations For The Authors):

Consider moving Figure 3 to be a supplement of Figure 4. This figure shows a water control and therefore best supplements the alanine experiment.

We would like to keep this figure as a main figure: we consider it very important to establish the validity of our behavioral setup at the beginning of the ms, and to establish that in all the following figures we are recording bona fide olfactory responses.

"sensory changes in mecano-sensory and gustatory systems" - mechano-sensory.

Done

Figure 2 legend: "(3) the right track is the 3D plus time (color-coded)" - shouldn't it be 2D plus time or 3D (x,y, time).

True! Thanks for noting this, corrected.

Figure 4 legend "E, Change in swimming patterns" should be H.

Done

"suggesting that their detection threshold for serine is lower than for alanine" - higher?

Done

In the behavioural plots, I assume that the "mean position" value represents the mean position along the X-axis of the chamber - this should be clarified and the axis label updated accordingly.

That is correct and has been updated in Methods and Figures and legends.

"speed, back and forth trips in X and Y, position and pattern changes (see Methods; Figure 7A)." - here it would be helpful to add an explanation like "to define an olfactory score for individual fish."

This has been changed in Results and more detailed explanations on score calculations are now given in Methods.

"possess enhanced mecanosensory lateral line" - mechanosensory.

Done

<https://doi.org/10.7554/eLife.92861.2.sa2>